nature genetics

Article

https://doi.org/10.1038/s41588-026-02579-x

Interpretable, flexible and spatially aware integration of multiple spatial transcriptomics datasets from diverse sources

Received: 17 October 2024

Accepted: 24 March 2026

Published online: 27 April 2026

Check for updates

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Recent advances in spatial transcriptomics (ST) have generated an expanding collection of heterogeneous datasets, offering unprecedented opportunities to investigate tissue organizations and functions. However, effective interpretation and integration of data originating from diverse sources and conditions remain a major challenge. We present INSPIRE, a deep-learning method for interpretable, integrative analysis of multiple ST datasets. INSPIRE adopts an adversarial learning strategy with graph neural networks to achieve spatially informed and adaptive data integration. By incorporating non-negative matrix factorization, INSPIRE identifies interpretable spatial factors and associated gene programs that characterize tissue architecture, cell-type organization and biological processes. Across a broad range of applications, INSPIRE demonstrates superior performance in resolving fine-grained biological signals, integrating complementary strengths across technologies, capturing condition-specific variation, uncovering tumor microenvironment heterogeneity, elucidating developmental dynamics and facilitating three-dimensional tissue reconstruction. INSPIRE also scales to extremely large datasets, as demonstrated by applications to Xenium-profiled human breast cancer and Stereo-seq mouse organogenesis datasets.

Spatial transcriptomic (ST) technologies enable transcriptome profiling with spatial context in intact tissues[1,2]. Diverse technologies with complementary strengths have emerged[3,4]. Sequencing-based methods (for example, Visium[5], Slide-seq[6,7], Stereo-seq[8], Visium HD[9]) support transcriptome-wide profiling at spot or subcellular resolution. While in situ hybridization (for example, seqFISH[10,11], MERFISH[12]) and in situ sequencing methods (for example, STARmap[13,14], Xenium[15]) require predefined gene panels, they can offer single-cell resolution by assigning transcripts to segmented cells. Together, these technologies provide great opportunities to dissect tissue architecture[16,17], cell–cell interactions[18,19] and spatial developmental trajectories[20,21].

Dimension reduction methods are widely used to extract biological patterns from high-dimensional transcriptomic data. Among them, non-negative matrix factorization (NMF) has proven an appealing approach[22–24]. In single-cell RNA-sequencing (scRNA-seq) analyses, NMF-based methods decompose expression into interpretable gene

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programs associated with cell identities and cellular activities[25,26], unraveling cell states under perturbations[27,28]. Recently, leveraging the decomposition nature of NMF[29], SpiceMix[30] and NSFH[31] adapts NMF for ST to model spatial dependence and decompose the data into interpretable spatial factors that encode distinct spatial patterns. While excelling at revealing spatial organization of cell identities, spatially variable features and biological processes, these methods are primarily designed for single or replicate sections and lack mechanisms to handle complex unwanted variation across sections. Therefore, developing interpretable, spatially aware methods that effectively integrate multiple diverse sections remains an important challenge.

Advances in ST have enabled numerous studies across technologies, often generating multiple sections, such as sagittal and coronal brain sections[32–34], and mouse-embryo sections across developmental time points[8,35]. Effectively interpreting these data is crucial for deciphering tissue architectures and their developmental dynamics. However, joint interpretation of diverse datasets is frequently confounded by unwanted variation across samples, technologies, conditions and time points[36,37]. Consequently, methods like SpiceMix and NSFH, while effective at identifying interpretable spatial factors in single or replicate sections, face difficulties in disentangling shared biological signals from heterogeneous unwanted variation. This difficulty is further amplified when certain datasets contain unique, biologically meaningful spatial factors[38]. Collectively, these challenges highlight the need for computational methods specifically designed for joint interpretation of multiple, diverse ST datasets.

Here we introduce INSPIRE, a deep-learning-based framework, that unifies NMF and adversarial learning[39] to achieve interpretable, flexible and spatially aware integration of ST datasets. INSPIRE models gene expression using a unified probabilistic framework that accounts for sequencing depths and represents expression profiles as products of harmonized spatial factors and shared gene loadings across sections. Spatial context is incorporated through graph neural networks (GNNs) that capture local cellular microenvironments, while a tailored adversarial learning strategy adaptively separates intrinsic biological variation from complex unwanted variation across datasets, even in the presence of dataset-specific biological signals. By integrating adversarial learning with NMF, INSPIRE enables harmonized discovery of interpretable spatial factors across multiple datasets. For these spatial factors, INSPIRE explicitly models their gene signatures, facilitating biological interpretation and identification of spatial gene programs.

We applied INSPIRE to a wide range of datasets, including human cortex sections from multiple donors, mouse brain sections at different orientations and resolutions, mouse skin sections under different conditions, human breast cancer sections with multiple tumor subtypes and mouse whole-embryo sections spanning organogenesis. Across these analyses, INSPIRE harmonizes diverse datasets, deciphers fine-grained spatial architecture, elucidates cell-population distributions and uncovers biological processes in tissues. INSPIRE supports diverse downstream analyses, including pathway enrichment analysis, spatially variable gene detection, spatial trajectory inference, gene expression imputation and three-dimensional (3D) tissue reconstruction. Notably, INSPIRE scales to large high-resolution datasets, as demonstrated in Xenium-profiled human breast cancer and Stereo-seq mouse organogenesis data, where it reveals tumor microenvironment heterogeneity and developmental dynamics.

Results

Method overview

INSPIRE takes gene expression count matrices and spatial coordinates from multiple sections as inputs. A spatially informed GNN-based encoder[40,41], $f_Z(\cdot)$, maps cells or spots from sections $s = 1, 2, \cdots, S$ into a shared latent space to generate their representations. To align these representations across sections while preserving section-specific variation, INSPIRE incorporates a tailored adversarial learning strategy[36,39] (Fig. 1a_i). Specifically, an auxiliary discriminator network, $D^s(\cdot)$, is deployed in the latent space to identify poor mixing between sections s and $s + 1$. Its feedback guides the encoder to improve the alignment. By introducing $S - 1$ discriminators, INSPIRE effectively harmonizes representations of cells or spots from all S sections in the latent space.

INSPIRE adopts an integrated NMF across sections to further decompose the biological signals in the latent space into spatial factors and gene programs (Fig. 1aii), characterizing tissue structures at a finer-grained level with enhanced interpretability. The integrated NMF includes harmonized non-negative spatial factor matrices across multiple sections, and a shared non-negative gene-loading matrix among sections. Spatial factors represent normalized enrichments of spatial patterns among cells or spots, while gene loadings explicitly encode gene importances underlying each pattern. Two designs enable effective multisection integrated NMF—first, a decoder network, $f_\beta(\cdot)$, generates spatial factors directly from integrated latent representations, where unwanted variations are eliminated, ensuring that factors reflect only biological variations. Second, section-specific and gene-specific effects are introduced to explicitly model residual confounding signals.

INSPIRE jointly optimizes the encoder, decoder and gene loadings under a unified objective. After training, it outputs integrated latent representations, spatial factors and interpretable gene loadings. These outputs support various downstream analyses (Fig. 1b). Details are included in the Methods.

INSPIRE offers superior accuracy and interpretability for the integrative analyses of multiple ST datasets

We conducted a simulation study to benchmark INSPIRE against existing methods using synthetic sections with known ground-truth spatial factors and gene loadings. Spatial factors were aligned with predefined domains, and varying levels of batch effects and noise were introduced to assess each method's ability to separate biological signals from unwanted variation (Supplementary Note 1 and Supplementary Figs. 1 and 2).

We first compared spot representations learned by INSPIRE with those from spatial NMF-based methods (SpiceMix[30], NSFH[31]), scRNA-seq integration methods (Harmony[42], Seurat[43], LIGER[24]) and spatial integration methods (PRECAST[37], MEFISTO[44], GraphST[45]). We also evaluated representation learning methods (STAGATE[46], BANKSY[47], SpaGCN[48], SpatialGlue[49]) followed by Harmony integration. Performance was evaluated using adjusted Rand index (ARI), normalized mutual information (NMI) and average silhouette width (ASW), which quantify agreement with ground-truth spatial domains and domain separability (Methods). As batch effects increased, interpretable methods SpiceMix and NSFH showed declining performance, reflecting their lack of explicit modeling for unwanted variation. In contrast, across all simulation scenarios, INSPIRE consistently achieved the best performance (Fig. 2a and Supplementary Figs. 3–6).

We then evaluated recovery of true spatial factors and their gene signatures to assess interpretability. For NMF-based (SpiceMix, NSFH, LIGER) and factor analysis-based methods (PRECAST, MEFISTO), factor and loading recovery were measured by Pearson correlation between estimated and true values. For other methods, factor recovery was assessed based on spot representations. Under weak batch effects and low noise, NMF-based approaches outperformed others, highlighting NMF's strength in disentangling true composite signals from high-dimensional data. INSPIRE consistently achieved the highest accuracy across different levels of batch effects and noise, demonstrating the robustness and interpretability advantages of joint spatial NMF across multiple sections (Fig. 2b and Supplementary Figs. 7–10).

Next, we applied INSPIRE to a Visium human dorsolateral prefrontal cortex (DLPFC) dataset[50] comprising 12 sections from three donors with annotated cortical layers (L1–L6) and white matter. When

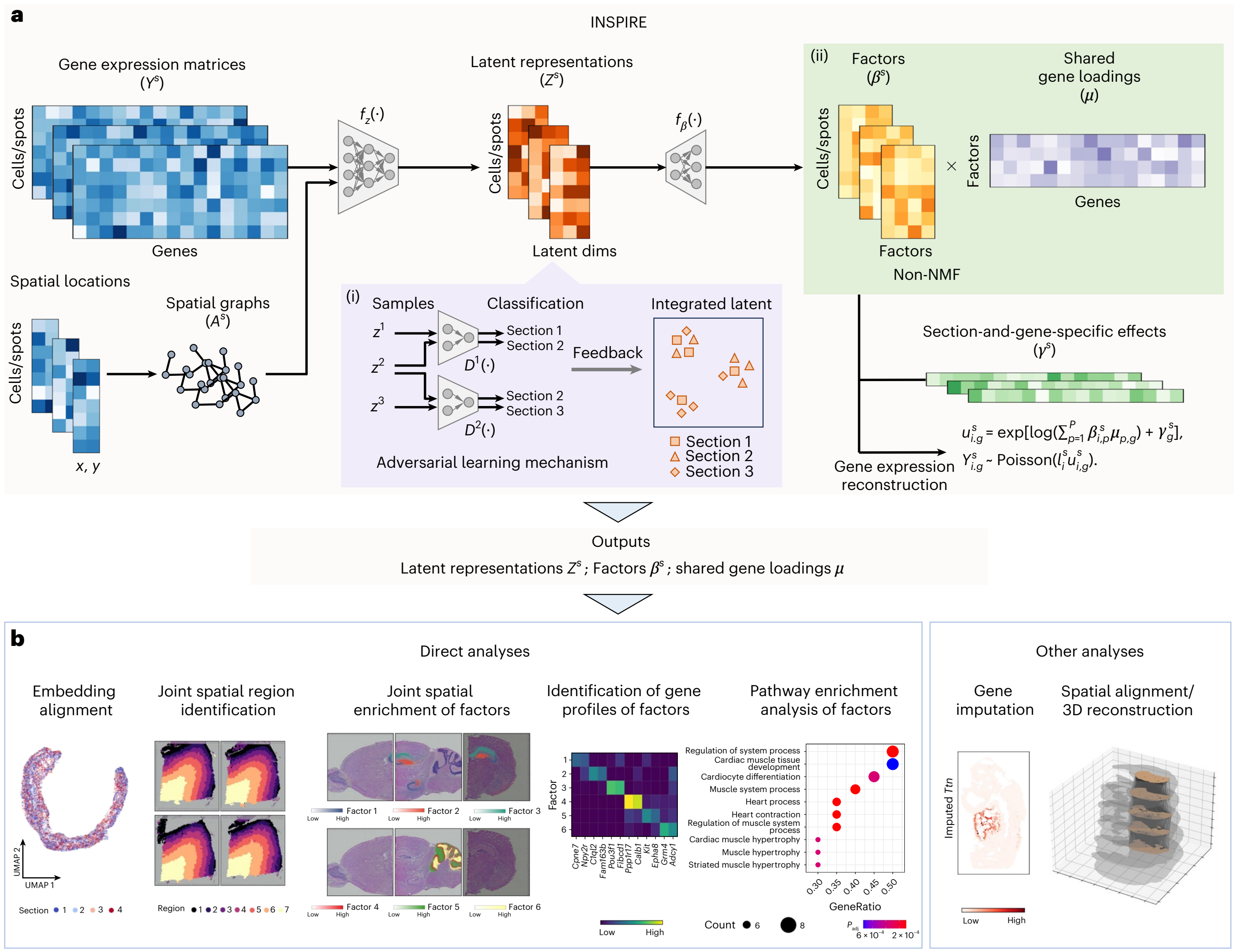


Fig. 1 | Overview of INSPIRE. a, INSPIRE is a unified deep-learning method that incorporates adversarial learning and NMF for interpretable integrations of ST datasets. Raw gene expressions and spatial locations from multiple sections are taken as the inputs. (i) INSPIRE embeds the biological variations from multiple sections into a shared latent space. By incorporating a tailored adversarial learning mechanism, INSPIRE effectively eliminates unwanted variations in this latent space, providing harmonized representations of cells or spatial spots among sections. (ii) The latent space enables INSPIRE to achieve an integrated NMF for multiple sections, further decomposing biological signals into a set of consistent and interpretable factors among sections. Unconfounded by unwanted variation, these spatial factors reveal detailed spatial organization across multiple tissue sections. The gene signatures of these spatial factors are explicitly characterized by the shared gene-loading matrix, thereby elucidating their biological meanings. After training, INSPIRE simultaneously outputs integrated latent representations, interpretable spatial factors and corresponding gene loadings. **b**, INSPIRE's outputs enable multiple downstream analyses, including spatial trajectory inference, identification of fine-grained spatial regions and tissue structures, detection of spatially variable genes and pathway enrichment analysis for deciphering biological processes in tissues. INSPIRE can also be applied to tasks such as gene imputation and 3D reconstruction of tissues from multiple parallel 2D sections. UMAP, Uniform Manifold Approximation and Projection.

integrating four sections 151673–151676 from the same donor, INSPIRE aligned them in the latent space while preserving layer separation. It revealed a shared L1–L6–white matter trajectory consistent with corticogenesis (Fig. 2d). Using these representations, INSPIRE accurately recovered cortical layers across sections, closely matching manual annotations (Fig. 2e,f).

Benchmarking spot representations in within-donor integration using ARI, NMI and ASW showed that INSPIRE achieved overall better performance than all competing methods, including domain-assignment approaches (BayesSpace[51], SpatialLeiden[52], BASS[53]) and PASTE[54] (Fig. 2c and Supplementary Figs. 11–15). In the more challenging across-donor integration of all 12 sections, INSPIRE again achieved better performance across all metrics. In contrast, scRNA-seq integration methods exhibited inferior performance due to the lack of spatial modeling. PRECAST showed improved performance but did not preserve the cortical trajectory, whereas SpiceMix and NSFH captured the trajectory but showed limited ability to integrate information across sections to further improve performance (Fig. 2g and Supplementary Figs. 16 and 17).

We note that the gained accuracy in the latent space is contributed by the NMF component in INSPIRE. For illustration, we ablated NMF to create 'INSPIRE (without NMF)' (Methods). Removing NMF led to consistently lower performance scores, demonstrating INSPIRE's effectiveness of integrating NMF with the latent space for learning accurate spot representations (Supplementary Fig. 18).

Finally, we evaluated the spatial factors and associated gene loadings inferred by INSPIRE. Evaluated by factor diversity and factor coherence metrics[55] (Methods), INSPIRE achieved higher-quality spatial factors than SpiceMix and NSFH in all within-donor integrations where all methods achieved reasonable alignment (Fig. 2h and

Supplementary Figs. 19–22). Visualization of factors, for example, in sections 151673–151676, revealed layer-specific enrichment (for example, L2, L5, L6), with gene loadings corresponding to known excitatory neuronal subtypes confirmed by an external scRNA-seq atlas (Fig. 2i–l). INSPIRE further identified factor-associated spatially variable genes (Methods) with validated reliability (Fig. 2m,n). INSPIRE also uncovered additional spatial organization not captured by manual annotations, such as the spatial distribution of astrocytes, with gene signature explicitly characterizing astrocytes' profile (Fig. 2k–n). This highlights INSPIRE's ability to decipher fine-grained and interpretable spatial structures through spatial factors and gene loadings.

Precise stitching of multiple sagittal and coronal mouse brain sections with partially shared spatial structures

Here we evaluate INSPIRE in a more challenging scenario—integrating multiple sections from different samples with only partially overlapping spatial structures. This task requires identifying shared, interpretable spatial factors while preserving section-unique biological signals and mitigating unwanted variation.

We applied INSPIRE to mouse brain sagittal anterior[32], sagittal posterior[33] and coronal sections[34] (Supplementary Fig. 23). Brain structures captured in these sections are only partially shared. INSPIRE aligned spot representations across sections in the latent space, where it partitioned the brain into 36 well-organized, coherent spatial regions. Among them, six layer-structured spatial domains formed the isocortex shared across sections, while section-unique structures, such as the cerebellum in the sagittal posterior section, and the main olfactory bulb in the sagittal anterior section were accurately preserved (Fig. 3a,b,d). In contrast, spatial NMF-based SpiceMix and NSFH, as well as PRECAST that performed relatively well in the benchmarking study, produced results confounded by strong batch effects and did not recover shared cortical layers across sections (Fig. 3e and Supplementary Figs. 24–27). Quantitative evaluation using the integration local inverse Simpson's Index (iLISI) metric (Methods) within the shared isocortex region confirmed superior cross-section alignment by INSPIRE (Fig. 3c).

We further verified that INSPIRE's discriminators (Methods) adaptively distinguish shared from section-unique biological signals. In the latent space, discriminators were active for shared regions such as the isocortex, promoting alignment, but inactive for section-unique structures such as the cerebellum, thereby preserving their identities (Supplementary Note 2, Fig. 3b and Supplementary Figs. 28–30).

Finally, we investigated the spatial factors inferred by INSPIRE. They captured fine-grained brain organization (Supplementary Fig. 31). For instance, distinct spatial factors delineated hippocampal neuron subtypes across CA1, CA2/CA3 and DG, with spatial distributions clearly separated and gene loadings concordant with subtype-specific markers from an external scRNA-seq atlas[56] (Fig. 3f,h,i and Supplementary Figs. 32 and 33). Using these factors, regional marker genes were successfully identified, including *Pou3f1*, *Fibcd1* for CA1; *Cpne7*, *Npy2r* for CA2/CA3; and *C1ql2*, *Fam163b* for DG (Fig. 3f,g). INSPIRE also identified neurons specific to molecular, Purkinje and granular layers in the cerebellum, which are unique to the sagittal posterior section, through distinct spatial factors (Fig. 3j). The associated gene signatures highlighted layer-specific marker genes (for example, *Kit*, *Epha8*; *Ppp1r17*, *Calb1*; *Grm4*, *Adcy1*), validated using Allen Brain Atlas in situ hybridization data[57] (Fig. 3k–m). In contrast, spatial factors inferred by SpiceMix and NSFH were confounded by batch effects, limiting their effectiveness in multisection analyses (Fig. 3c,e and Supplementary Figs. 34 and 35). Together, these results demonstrate INSPIRE's advantage in integrating partially overlapping sections to construct a comprehensive tissue atlas and in revealing interpretable, fine-grained spatial architecture with clear biological meaning.

INSPIRE integrates ST data from different technologies, facilitating multiple downstream analyses

Here we demonstrate that INSPIRE can integrate data across technologies by leveraging their complementary strengths to enhance biological insights. We analyzed mouse brain datasets generated by Slide-seqV2 (ref. 7) and MERFISH[58]. Slide-seqV2 profiled over 20,000 genes at near-cellular resolution, whereas MERFISH measured expression at single-cell resolution but covered only 1,122 genes. After standard Scanpy preprocessing, Slide-seqV2 failed to recover known isocortical layers, while MERFISH clearly revealed such structure (Fig. 4a and Supplementary Fig. 36). Integrating these datasets is challenging due to strong technical discrepancies (Supplementary Fig. 37) but enables combining MERFISH's higher data quality for accurate spatial annotation with Slide-seqV2's transcriptome-wide coverage.

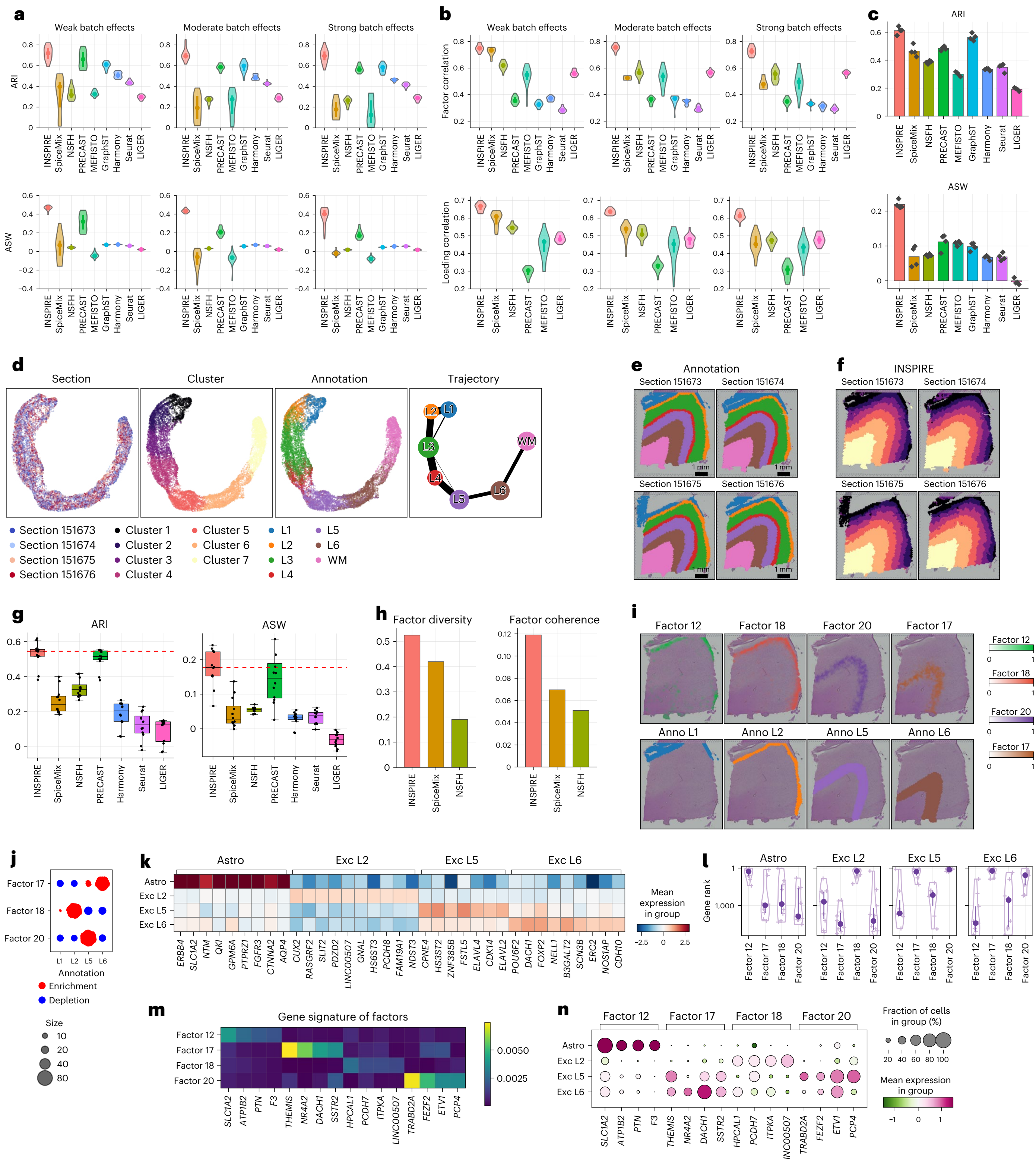
INSPIRE effectively removed technical confounders and aligned spatial regions across datasets. It identified major hippocampal subregions and broad cortical layers, and its spatial factors further revealed finer-grained cortical layers consistent across datasets (Fig. 4b,e and Supplementary Fig. 38). Notably, INSPIRE called out cortical layers in the Slide-seqV2 data by leveraging information from MERFISH (Fig. 4a,b), thereby enabling downstream analyses that were not possible using MERFISH alone. For instance, MERFISH did not capture gene *Plxna2*. However, after integration, COMMOT[18] identified the Sema6A–Plxna2 signaling interaction across INSPIRE's annotated regions in the Slide-seqV2 data (Fig. 4f,g), suggesting potential cell migration and neuronal connectivity around inner cortical layer area[59]. In contrast, SpiceMix, NSFH and PRECAST produced representations and spatial factors that remained confounded by technical effects. Consistent with this, quantitative evaluation using iLISI and cell-type

Fig. 2 | Benchmarking of INSPIRE and state-of-the-art methods based on simulated data and the human DLPFC dataset. a, ARI and ASW scores of the benchmarked methods evaluated on simulated tissue sections with varying strengths of batch effects. Each violin plot is based on 20 replicate experiments. Dots indicate median values, and bars indicate the range between the 25th and 75th percentiles, calculated based on $n = 20$ points. **b**, Pearson's correlation between the estimated and ground-truth values of spatial factors and gene loadings for the benchmarked methods under varying levels of batch effects in the simulation study. Each violin plot is based on 20 replicate experiments. Dots indicate median values, and bars indicate the range between the 25th and 75th percentiles, calculated based on $n = 20$ points. **c**, ARI and ASW scores of the benchmarked methods evaluated on four human DLPFC sections (indices 151673–151676; $n = 4$). **d**, UMAP plots of spot representations from INSPIRE, colored by section indices, INSPIRE's assigned spatial domain labels and manual annotations. The PAGA was applied to the spot representations for spatial trajectory inference. **e**, Manual annotations on sections 151673–151676. **f**, Spatial domain identification result from INSPIRE. **g**, ARI and ASW scores of the compared methods assessed on all 12 DLPFC sections ($n = 12$) from three donors. Each box plot ranges from the first to third quartiles with the median as the horizontal line, while whiskers represent 1.5× the interquartile range from the lower and upper bounds of the box. The dashed line indicates the median score of INSPIRE. **h**, Factor diversity and factor coherence scores of the benchmarked methods on human DLPFC sections 151673–151676. **i**, Spatial distributions of factors 12, 18, 20 and 17 identified by INSPIRE on section 151673. **j**, Enrichment or depletion of different spatial factors in cortical layers. **k**,**l**, Identifying marker genes for the four cell types using a scRNA-seq atlas by selecting genes ranked in the top 10 by Scanpy and with fold changes greater than 1.5 relative to other cell types (**k**), and visualizing the rank distribution of them in different spatial factors (**l**; $n = 9$ for astrocytes, $n = 10$ for excitatory neurons in layer L2, $n = 7$ for excitatory neurons in layer L5, and $n = 9$ for excitatory neurons in layer L6). Dots indicate median values, and bars indicate the range between the 25th and 75th percentiles. **m**, Spatial factor-specific genes identified by INSPIRE. **n**, Expression levels of the factor-specific genes among cell types. WM, white matter; PAGA, partition-based graph abstraction algorithm; Astro, astrocytes; Exc, excitatory neuron.

local inverse Simpson's Index (cLISI) in the hippocampus (Methods) indicated their inferior integration and weaker preservation of biological signals (Fig. 4c,d and Supplementary Figs. 39–43).

We further applied INSPIRE to analyze mouse whole-embryo sections generated by seqFISH[35] and Stereo-seq[8]. Despite strong discrepancies, INSPIRE aligned cell embeddings and corrected cross-platform expression differences (Supplementary Note 3, Fig. 4h and Supplementary Figs. 44–46), enabling identification of 16 consistent spatial domains, including midbrain and developing heart regions. INSPIRE's spatial factors further characterized fine-grained cardiac structures and depicted biological processes, such as angiogenesis and vasculature development (Fig. 4j,k and Supplementary Figs. 47 and 48).

INSPIRE's integrative framework also enabled the accurate transfer of information across technologies. By holding out marker genes from seqFISH and imputing their spatial expression using Stereo-seq, we obtained patterns consistent with measured expression. We also confirmed robust imputation accuracy across multiple metrics and cross-validation schemes (Supplementary Note 4, Fig. 4l and



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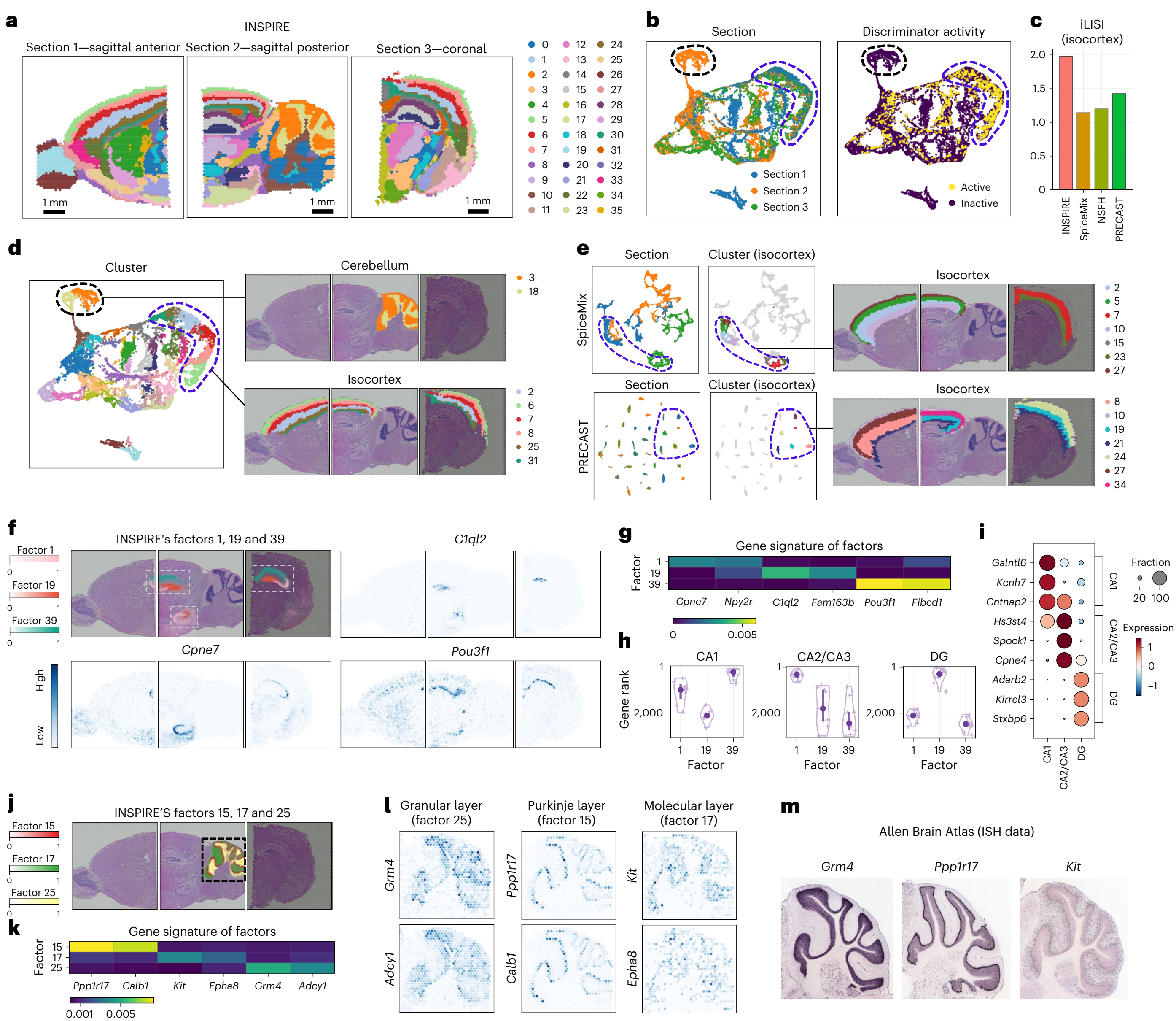

Fig. 3 | Analysis of multiple mouse brain sections with only partially shared spatial tissue organizations. INSPIRE integrated the sagittal anterior, sagittal posterior and coronal sections of the mouse brains. **a**, Spatial clusters visualized on the three tissue sections. **b**, UMAP plots of spot representations from INSPIRE, colored by section indices and discriminator activities. INSPIRE's discriminators were active on the spots describing the shared spatial structures across sections, while they were inactive on spots related to the section-unique structures. Thereby, they adaptively guided INSPIRE to align shared variations among sections, while preserving section-unique signals. **c**, Comparison of iLISI scores evaluating integration performance among benchmarked methods. Higher iLISI scores indicate better alignment performance. **d**, Spatial regions 3 and 18 from INSPIRE characterized the cerebellum. Spatial regions 2, 6, 7, 8, 25 and 31 characterized layers in the isocortex. **e**, UMAP plots of spot representations from SpiceMix and PRECAST (left), and visualizations of their spatial clusters in the isocortex (right). **f**,**g**, Spatial distributions of factors 1, 19 and 39 identified by INSPIRE (**f**), based on the learned gene signatures (**g**), INSPIRE identified spatially variable genes associated with the three factors, respectively. White dashed squares highlight the hippocampus. **h**,**i**, Identifying marker genes of CA1, CA2/CA3 and DG using a scRNA-seq atlas by selecting genes ranked in the top ten by Scanpy and with fold changes greater than 1.5 relative to other cell populations (**i**), and visualizing the rank distribution of them in gene loadings of the three factors (**h**; $n = 7$ for CA1, $n = 9$ for CA2/CA3, and $n = 10$ for DG). Dots indicate median values, and bars indicate the range between the 25th and 75th percentiles. **j**,**k**, Spatial distributions (**j**) and gene signatures (**k**) of factors 5, 17 and 25, respectively. The black dashed square indicates the cerebellum. **l**, Top spatially variable genes associated with each of the three factors identified by INSPIRE. **m**, In situ hybridization (ISH) images of genes *Grm4*, *Ppp1r17* and *Kit* from Allen Brain Atlas.

Supplementary Figs. 49–56). Extending this approach, we imputed genes from Stereo-seq into seqFISH, expanding coverage from 351 to over 20,000 genes. This enabled differential expression analysis between dorsal and ventral regions of the developing gut tube, which were clearly separated only in the seqFISH data (Fig. 4m). We identified dorsal-enriched genes (*Smoc2*, *Tbx1*, *Cdh2*, *Chrd*, *Foxd1*) and ventral-enriched genes (*Tbx3*, *Id1*, *Isl1*, *Dlk1*, *Pmp22*, *Nkx2-3*, *Gata3*). These results are consistent with known dorsal–ventral gut tube patterning, with esophagus-associated genes (for example, *Foxd1*) enriched dorsally and trachea-associated genes (for example, *Tbx3*, *Id1*, *Isl1*) enriched ventrally. Notably, *Id1*, *Pmp22* and *Foxd1* were imputed (Fig. 4m,n), underscoring INSPIRE's ability to enable biological discoveries. Applying SpiceMix, NSFH and PRECAST to this cross-technology setting again resulted in inferior integration and lower iLISI scores

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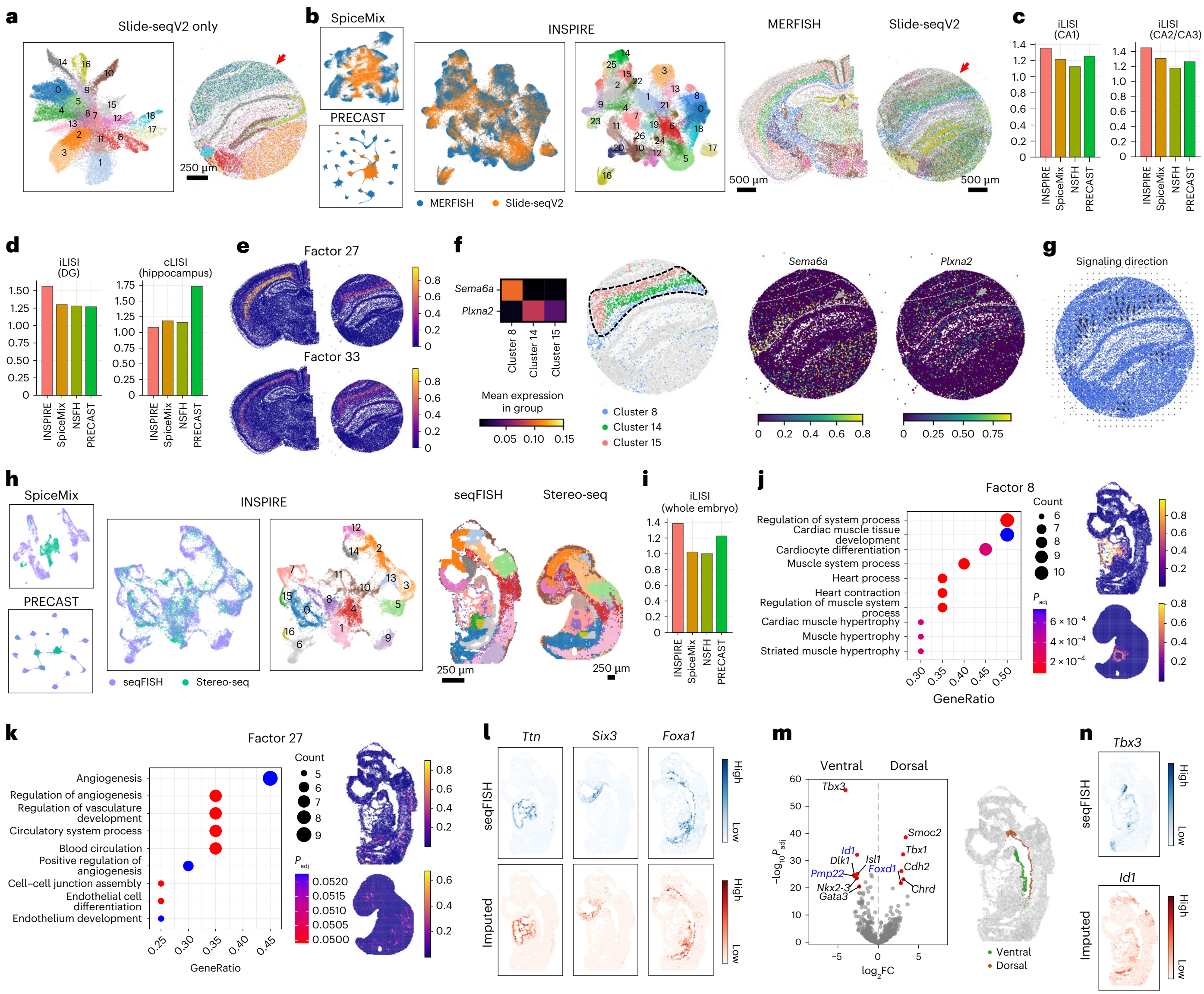

Fig. 4 | Integrative analyses of data from different ST technologies to enhance biological insights. a, Spatial domain identification from the analysis of Slide-seqV2 data alone. The red arrow highlights the isocortex region. **b**, Cell representations aligned between the MERFISH and Slide-seqV2 datasets using SpiceMix, PRECAST, INSPIRE and spatial domain identification result from INSPIRE. **c**,**d**, Comparison of iLISI and cLISI scores among methods. **c**, iLISI scores were evaluated in CA1 and CA2/CA3. **d**, iLISI scores were assessed in DG, and cLISI scores were measured in the hippocampus. A higher iLISI score indicates better data alignment performance, while a lower cLISI score represents better preservation of biological variation. **e**, Spatial distributions of spatial factors 27 and 33. **f**, Expressions of genes *Sema6a*, *Plxna2* among INSPIRE's identified spatial regions 8, 14 and 15. **g**, Signaling direction from ligand Sema6A to receptor Plxna2 across INSPIRE's annotated regions inferred by the COMMOT method. **h**, Cell representations aligned between the seqFISH and Stereo-seq embryo datasets using SpiceMix, PRECAST, INSPIRE and spatial domain identification result from INSPIRE. **i**, iLISI scores across the embryo sections among methods. **j**,**k**, Significant GO terms and spatial distributions of INSPIRE's factors 8 (**j**) and 27 (**k**). GO enrichment was performed using the clusterProfiler package, with *P* values computed using one-sided hypergeometric tests and multiple-testing correction using the Benjamini–Hochberg procedure. **l**, Manually held out genes from the seqFISH dataset and imputed their expressions from the Stereo-seq dataset and visualized the expressions of *Ttn*, *Six3* and *Foxa1* measured by seqFISH (top), and compared them with the imputed expression levels (bottom). **m**, Differentially expressed gene analysis between the ventral and dorsal sides of the developing gut tube in the seqFISH dataset. Top differentially expressed genes with imputed expression levels from INSPIRE are annotated in blue, and top genes with expression levels quantified by seqFISH are annotated in black. **n**, *Tbx3* expression levels measured by seqFISH (top), and *Id1* expression levels imputed by INSPIRE (bottom). FC, fold change.

(Fig. 4h,i and Supplementary Figs. 57–61). Together, these analyses show INSPIRE's broad applicability.

INSPIRE facilitates the characterization of tumor microenvironment heterogeneity in human breast cancer

We applied INSPIRE to mouse skin wound-healing data[60], demonstrating its ability to capture condition-specific spatial organization. INSPIRE jointly analyzed uninjured and postoperative day 7 wound sections, aligning shared skin structures while preserving wound-specific signals. In the latent space, aligned spots formed a continuous trajectory that reflected the layered skin architecture, whereas wound-specific spots near the injury center remained distinct. In contrast, SpiceMix showed less effective alignment across sections. While NSFH and PRECAST aligned shared layers, their detected layers show reduced spatial continuity, as confirmed by higher spatial discontinuity scores (Methods; Supplementary Figs. 62 and 63).

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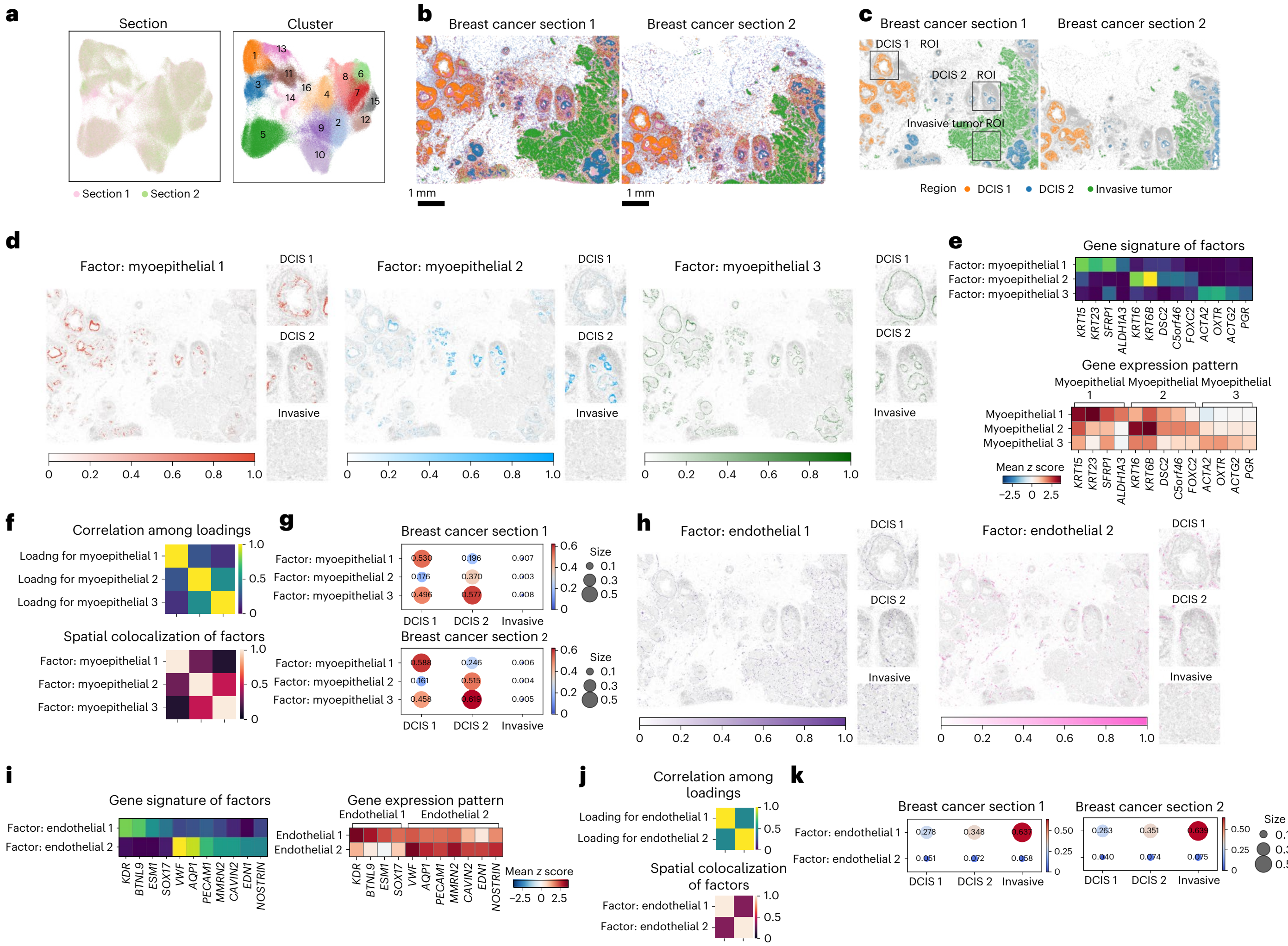

Fig. 5 | Analysis of human breast cancer Xenium sections to reveal heterogeneity within the tumor microenvironment. a, UMAP plots of spot representations, colored by section indices and spatial domain labels assigned by INSPIRE. **b**, Visualization of the identified spatial domains across sections. **c**, Highlighted ROIs annotated by the pathologist, including DCIS 1, DCIS 2 and an invasive tumor region, each corresponding to a distinct cluster identified by INSPIRE. **d**, Spatial distributions of three inferred factors, each representing a distinct myoepithelial subpopulation. **e**, Gene signatures of the three myoepithelial-related factors inferred by INSPIRE, and gene expression pattern among these factor-associated cells. **f**, Pearson's correlation among the gene loadings of the three myoepithelial factors, and spatial colocalization patterns of the three factors. **g**, Enrichment of cells associated with the three myoepithelial factors within the tumor microenvironments of distinct subtypes across both sections. **h**, Spatial distributions of two inferred factors, each representing a distinct endothelial subpopulation. **i**, Gene signatures of the two endothelial-related factors inferred by INSPIRE, and gene expression pattern among these factor-associated cells. **j**, Pearson's correlation among the gene loadings of the two endothelial factors, and spatial colocalization patterns of the two factors. **k**, Enrichment of cells associated with the two endothelial factors within the tumor microenvironments of distinct subtypes across both sections. ROIs, regions of interest.

The inferred spatial factors provide insights into INSPIRE's better ability to depict detailed spatial regions. Two factors captured epidermal spinous and basal layers, associated with known marker genes (*Krt1*, *Krt10*; *Krt5*, *Klf5*). Their biological relevance was further supported by their correct spatial localization[61] and by the loss of these factors near the wound center. As another example, INSPIRE identified a wound-enriched factor associated with macrophage marker genes (*Cd14*, *Itgam*), with Gene Ontology (GO) analysis highlighting immune and inflammatory processes specific to the wound environment, demonstrating its ability to model condition-specific signals (Supplementary Fig. 64).

Next, we applied INSPIRE to two high-resolution Xenium-profiled human breast cancer sections[15] from a stage II-B ER$^+$, HER2$^+$ tumor comprising over 280,000 cells. INSPIRE completed the analysis in 8 min, demonstrating scalability.

INSPIRE identified 15 clusters in the latent space. Three of them corresponded to pathologist-annotated tumor domains, including two ductal carcinoma in situ regions (DCIS 1 and DCIS 2) and one invasive tumor region (Fig. 5a–c). The identification of these regions was supported by expression patterns—all tumor regions showed high *ERBB2* expression, whereas elevated *ESR1* expression was specific to DCIS. INSPIRE delineated additional structured regions. For example, it marked myoepithelial layers enriched for *KRT14* and *KRT5* expression, which form layered structures in DCIS but absent in the invasive tumor (Supplementary Fig. 65). This highlights a key microenvironmental distinction between noninvasive and invasive tumor subtypes, consistent with the evidence that loss of the myoepithelial layer is a hallmark of progression from in situ to invasive breast cancer[62–64].

Spatial factor analysis further revealed fine-grained heterogeneity within the tumor microenvironment. INSPIRE identified three myoepithelial factors with distinct spatial localizations and gene signatures (*KRT15*, *KRT23*; *KRT16*, *KRT6B*; *ACTA2*, *OXTR*; Fig. 5d–f and Supplementary Fig. 66). Quantitative spatial proximity analysis showed different enrichments of these subpopulations near the two DCIS regions (Fig. 5g). Indicated by previous studies[65–67], enrichment

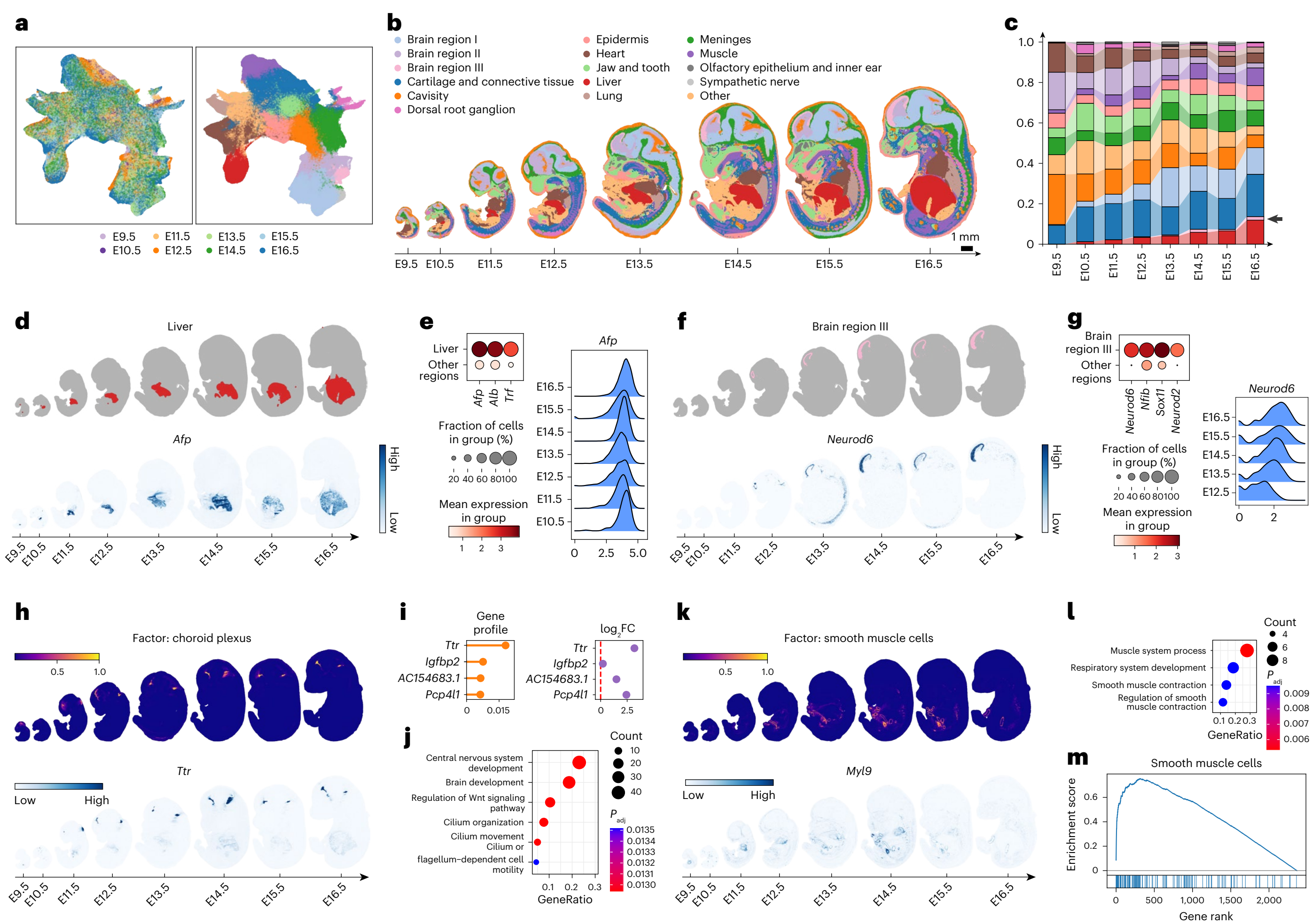

Fig. 6 | Construction of the spatiotemporal mouse organogenesis atlas by integrating whole-embryo sections across various developmental stages. **a**, UMAP plots of spot representations, colored by section indices and spatial domain labels assigned by INSPIRE. **b**, Visualization of the identified spatial domains across sections. **c**, For each developmental stage, we visualized the proportions of the embryo occupied by different spatial regions. The arrow highlights a small spatial region, brain region III (**f**), which was absent in early developmental stages but emerged in later stages. **d**,**e**, Visualization of the liver region (**d**) and the expression patterns of liver marker genes (**e**). **f**,**g**, Visualization of a brain region (**f**) and the expression pattern of its differentially expressed genes (**g**). **h**,**i**, Spatial distribution of the factor representing the choroid plexus (**h**) and its gene signature (**i**). The gene profile indicates the expression levels of genes for this factor. The $\log_2$FC measures the difference between gene expressions in this factor and other factors. **j**, GO analysis for the genes specific to the factor in **h**. GO enrichment was performed using the clusterProfiler package, with *P* values computed using one-sided hypergeometric tests and multiple-testing correction using the Benjamini–Hochberg procedure. **k**, Spatial distribution of the factor related to smooth muscle cells and the spatial expression pattern of a smooth muscle cell marker gene. **l**, GO analysis for the genes specific to the factor in **k**. GO enrichment was performed using the clusterProfiler package, with *P* values computed using one-sided hypergeometric tests and multiple-testing correction using the Benjamini–Hochberg procedure. **m**, Gene set enrichment analysis comparing genes related to the factor in **k** with the marker gene set for smooth muscle cells.

of *KRT16*-high and *KRT15*-low myoepithelial population near DCIS 2 region suggests a more aggressive phenotype, highlighting fine-grained differences among noninvasive tumor regions. As another example, INSPIRE identified two endothelial spatial factors with distinct transcriptional profiles (*KDR*, *BTNL9*; *VWF*, *AQP1*) and spatial distributions (Fig. 5h–j and Supplementary Fig. 67). Endothelial cells enriched for *KDR* and *ESM1* expression were preferentially localized near the invasive tumor relative to DCIS (Fig. 5k), consistent with the established roles of KDR/VEGFR-2 and ESM1 in tumor angiogenesis, progression and metastasis[68,69]. Together, these results demonstrate INSPIRE's ability to resolve subtle, biologically meaningful cellular heterogeneity within the tumor microenvironment.

INSPIRE creates a comprehensive mouse organogenesis atlas, enabling spatiotemporal analysis of embryonic development

To study spatiotemporal organogenesis, we applied INSPIRE to eight mouse whole-embryo Stereo-seq sections spanning embryonic day E9.5–E16.5 (ref. 8). Following the original study, data were binned to 50 × 50 DNA nanoballs (~25 μm), yielding over half a million spatial spots.

INSPIRE efficiently aligned all eight sections in the latent space, despite large sample sizes and varying embryo morphology. This integration identified consistent spatial regions corresponding to major organs, including heart, liver, lung and brain, validated by canonical marker genes across developmental stages (Fig. 6a,b and Supplementary Fig. 68). Using these annotated regions, we quantified organ-level dynamics across development. The liver expanded rapidly from an initially small domain, whereas the heart was already well-formed at early stages and remained relatively stable, consistent with its early functional role[70] (Fig. 6c–e and Supplementary Fig. 69). INSPIRE also captured tissues that were barely formed earlier but well-developed later. For example, the lung showed clear shape after E12.5 (Supplementary Fig. 70), consistent with the onset of branching morphogenesis at this stage[71]. INSPIRE also identified a forebrain

https://doi.org/10.1038/s41588-026-02579-x

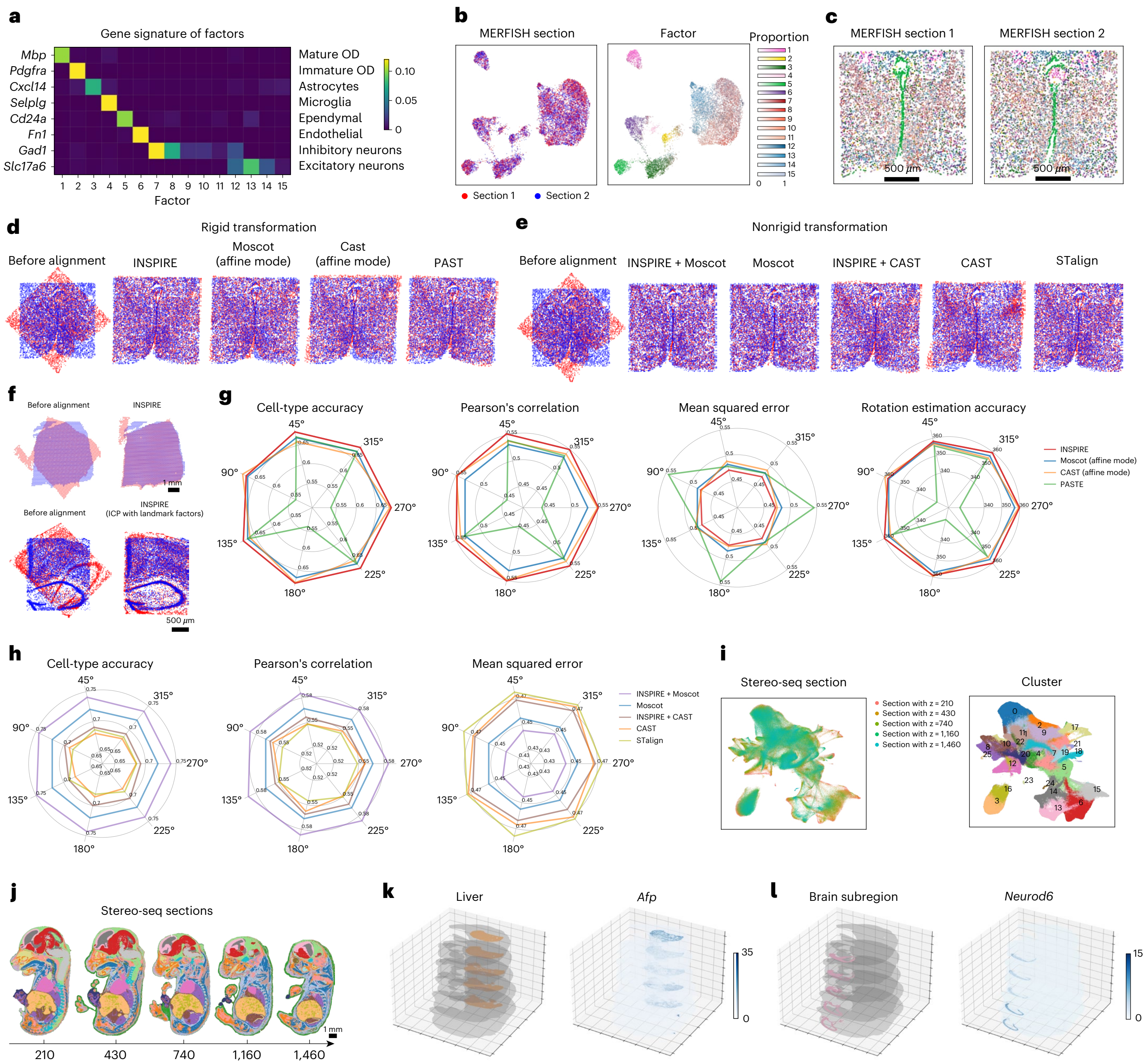

Fig. 7 | Three-dimensional reconstruction of tissues and whole organisms by INSPIRE. We applied INSPIRE to spatially align two MERFISH mouse hypothalamic preoptic sections (**a**–**e**), two Visium DLPFC sections, two STARmap PLUS mouse hippocampus sections (**f**) and five whole-embryo sections produced by Stereo-seq (**i**–**l**). **a**, Gene signatures of spatial factors, with each row indicating a marker gene specific to a cell type. **b**, UMAP plots of cell representations from INSPIRE, colored by MERFISH section indices and the inferred factor proportions among cells. **c**, The MERFISH sections colored by the proportions of factors. **d**,**e**, Comparison of spatial alignment results among rigid transformation-based methods (**d**) and nonrigid transformation-based methods (**e**). **f**, The Visium sections and the STARmap PLUS sections before and after INSPIRE's spatial alignment, respectively. **g**,**h**, Cell-type accuracy, Pearson's correlation, MSE and rotation estimation accuracy scores among rigid transformation-based methods (**g**) and nonrigid transformation-based methods (**h**) evaluated based on the MERFISH dataset. Higher cell-type accuracy, Pearson's correlation, rotation estimation accuracy and lower MSE reflect better performance. The rotation estimation accuracy is calculated as 360 minus the difference between the estimated and ground-truth rotation angles. **i**,**j**, UMAP plot of spot representations (**i**) and visualization of the Stereo-seq whole-embryo sections colored by the identified spatial domains (**j**). **k**, Visualization of liver structure and distribution of *Afp* expression on the reconstructed 3D model of the embryo. **l**, Visualization of the 3D structure of a brain subregion and the distribution of *Neurod6* expression on the reconstructed 3D embryo. MSE, mean square error.

region enriched for *Neurod6* and *Neurod2* since E12.5, suggesting active neuronal differentiation and development (Fig. 6f,g).

Beyond major organs, INSPIRE's NMF-based modeling uncovered fine-grained tissue architecture dynamics through spatial factors. For example, INSPIRE identified two factors in forebrain, whose spot abundance increased after E11.5–E12.5, revealing an emerging layered organization by E13.5. Similarly, three factors captured progressive development of the embryonic mouth and jaw, with two emerging at E11.5–E12.5 and one present across stages, together reflecting increasing structural complexity (Supplementary Figs. 71 and 72). Linking spatial factors to gene signatures further uncovered developmental programs. INSPIRE identified a choroid plexus factor enriched for processes including brain development, cilium organization and movement. Its spot count increased over time while its relative proportion decreased, suggesting limited scaling with overall embryo growth (Fig. 6h–j and Supplementary Fig. 73). As another example, a factor enriched around tube-like structures was detected, whose spatial distribution became pronounced after E12.5, consistent with sharp increases in both spot number and proportion after E11.5. Gene set enrichment against smooth muscle markers and its colocalization with *Nkx2-3* together suggested that this factor described nuanced distributions of gastrointestinal smooth muscle cells across development (Fig. 6k–m and Supplementary Figs. 74 and 75). Together, these results highlight INSPIRE's ability to uncover biologically meaningful spatiotemporal patterns during organogenesis through interpretable factors and gene profiles.

INSPIRE enables accurate 3D reconstruction of tissues and whole organisms

Recent ST studies increasingly generate parallel two-dimensional (2D) sections, enabling 3D reconstruction through joint analysis. INSPIRE can reliably register adjacent sections and support accurate 3D reconstruction across scales, from tissue regions to whole organisms.

We first evaluated INSPIRE on adjacent MERFISH sections from the mouse hypothalamic preoptic region[72]. INSPIRE identified interpretable spatial factors corresponding to cell types or subtypes and produced consistent factor distributions across sections, facilitating precise integration (Supplementary Note 5, Fig. 7a–c and Supplementary Fig. 76). Using mutual nearest neighbor cells as anchors, INSPIRE estimated optimal rigid transformations to align sections. With the MERFISH replicates, we created benchmark datasets by fixing one section and rotating the other at varying angles. Across multiple metrics, including cell-type accuracy, Pearson's correlation, mean squared error and rotation estimation accuracy, INSPIRE achieved an overall best performance compared to PASTE[54], Moscot[73] (affine mode) and CAST[74] (affine mode; Fig. 7d–g and Supplementary Fig. 77). Similarly, INSPIRE achieved strong alignment performance on benchmark datasets generated by Visium and STARmap PLUS (Supplementary Note 6 and Supplementary Figs. 78 and 79).

Next, we show that INSPIRE-derived representations enhance nonrigid spatial alignment. Incorporating these representations as inputs to representation-based nonrigid alignment tools Moscot and CAST results in hybrid approaches 'INSPIRE + Moscot' and 'INSPIRE + CAST', respectively. These approaches consistently enhanced nonrigid alignment accuracy compared to their original versions in all three benchmark datasets under all evaluation metrics. INSPIRE + CAST also led to better performance compared to recently developed method STalign[75] (Supplementary Note 6, Fig. 7h and Supplementary Figs. 80 and 81). These results highlight the utility of INSPIRE embeddings for improving spatial alignment.

Finally, we applied INSPIRE to five adjacent Stereo-seq mouse-embryo sections at E16.5, comprising over 560,000 spatial spots. INSPIRE efficiently integrated and sequentially aligned the sections, enabling 3D reconstruction of major organs such as the heart and liver (Fig. 7i–k and Supplementary Figs. 82 and 83). In addition, it revealed fine-grained tissue substructures, such as forehead subregions varying along the left-right axis (Fig. 7l and Supplementary Fig. 84). In summary, INSPIRE enables reliable 3D reconstruction of tissues and whole organisms, facilitating 3D spatial analysis beyond traditional 2D studies.

Discussion

We introduced INSPIRE for integrative and interpretable analysis of ST datasets across samples, technologies, biological conditions and developmental stages. Across simulations and diverse real-data applications, INSPIRE effectively integrated heterogeneous data and resolved fine-grained spatial architecture through interpretable spatial factors and associated gene programs.

Our analyses illustrate limitations of existing methods for multisection ST integration. scRNA-seq integration methods (Harmony, Seurat, LIGER) do not model spatial dependencies, leading to suboptimal performance. Methods that incorporate spatial modeling (PRECAST, GraphST, MEFISTO) showed limited ability to handle heterogeneous unwanted variation—PRECAST relies on a shared Gaussian mixture model that is less effective under strong section-specific effects, GraphST requires prior spatial registration and MEFISTO exhibits weaker integration with high computational cost. Recent spatial NMF-based methods (SpiceMix, NSFH) capture fine-grained spatial architecture but are largely restricted to single or replicate sections due to the lack of mechanisms for correcting complex unwanted variation. In contrast, INSPIRE overcomes these limitations through innovations in model design.

INSPIRE offers three key methodological advantages. First, its tailored adversarial learning adaptively separates unwanted variation from biological signals, even in the presence of section-specific biology, while the flexibility of neural networks enables correction of heterogeneous unwanted effects from diverse sources. Second, by integrating adversarial learning with NMF, INSPIRE extracts consistent and interpretable spatial factors across sections, unconfounded by complex unwanted effects. These factors capture and reveal fine-grained spatial architecture, support biological interpretation and improve spot or cell representations. Third, INSPIRE incorporates GNNs to model local cellular microenvironments, providing spatial awareness while maintaining scalability through lightweight, mini-batch-optimized networks, as demonstrated in large-scale applications and scalability analyses (Supplementary Notes 7 and 8 and Supplementary Figs. 85–89).

INSPIRE demonstrated broad applicability across challenging settings, enabling spatial atlas construction, cross-technology integration, discovery of condition-specific biology, resolution of tumor microenvironment heterogeneity, characterization of developmental dynamics and 3D tissue reconstruction. Despite its strengths, INSPIRE currently relies on genes shared across datasets for integrative interpretation, potentially excluding dataset-specific signals. Preliminary analyses suggest that incorporating nonshared genes could further improve integration (Supplementary Note 9 and Supplementary Figs. 90 and 91). In addition, although uncertainty in gene imputation can be quantified using conformal prediction (Supplementary Note 10 and Supplementary Fig. 92), uncertainty quantification for data integration remains an open challenge. Extensions such as Bayesian neural networks may address this limitation, but at the cost of increased model complexity.

As ST continues to grow in scale and diversity, scalable and interpretable integration of heterogeneous datasets will become increasingly essential. With strong performance, interpretability and versatility, INSPIRE provides a powerful addition to the ST analysis toolkit.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41588-026-02579-x.

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Methods

Model of INSPIRE

INSPIRE offers interpretable and spatially informed integration of ST data from multiple tissue sections. Let $s = 1, 2, \cdots, S$ be the index of tissue sections. For section s, we observe gene expression count matrix $Y^s = (y^s_{i,g}) \in \mathbb{R}^{n_s \times G}$ and 2D spatial coordinates of cells or spatial spots $P^s \in \mathbb{R}^{n_s \times 2}$, where $i = 1, 2, \cdots, n_s$ is the index for cells or spatial spots, and $g = 1, 2, \cdots, G$ is the index for genes. Using all the information as input, INSPIRE learns to decipher spatial structures across all the S sections and infers the associated gene programs.

To integrate both gene expressions and spatial locations across all tissue sections, INSPIRE encodes the gene expression information of all cells or spatial spots into a shared latent space $\mathcal{Z}$, using a neural network that accounts for spatial dependencies among cells or spots. Specifically, INSPIRE builds a 2D neighborhood graph for each section using the spatial location matrix. We denote the neighborhood graph for section s as $A^s = (a_{i,j}) \in \{0,1\}^{n_s \times n_s}$, where $a_{i,j} = 1$ if cells or spots i and j are spatial neighbors, and $a_{i,j} = 0$ otherwise. Using both gene expressions $\{Y^s\}_{s=1,2,\cdots,S}$ and spatial graphs $\{A^s\}_{s=1,2,\cdots,S}$, the latent representations of cells or spatial spots are generated by:

$$x^s_{i,g} = \log\left(\frac{y^s_{i,g}}{\sum_{g=1}^{G} y^s_{i,g}} M + 1\right), \tag{1}$$

$$Z^s = f_Z(X^s, A^s), \tag{2}$$

where we perform log-normalization on count matrices $\{Y^s\}_{s=1,2,\cdots,S}$ for algorithm stability. In the data normalization, we set $M = 10^4$ for sequencing-based ST data, and set $M = 10^3$ for imaging-based ST data. The log-normalized data $X^s = (x^s_{i,g}) \in \mathbb{R}^{n_s \times G}$ are then encoded to latent representations $Z^s \in \mathbb{R}^{n_s \times P}$ through a GNN $f_Z(\cdot)$ with parameters shared among all sections, where P is the dimensionality of the shared latent space. For section s, the latent representation Z^s embeds information from both gene expressions Y^s and spatial neighbors encoded in graph A^s, describing spatially informed biological variations in section s. Notably, in addition to capturing spatially aware biological signals in each section, the shared latent space is designed to achieve integration across all input sections. For harmonizing latent representations $Z^1, Z^2, \cdots, Z^S$ from different tissue sections, INSPIRE adopts a tailored adversarial mechanism in latent space $\mathcal{Z}$. Let $\mathbf{z}^s_i \in \mathbb{R}^P$ be the latent representation of cell or spot i in section s. An auxiliary discriminator network, $D^s(\cdot): \mathcal{Z} \to (0,1)$, is deployed to identify where the poor mixing between representations $\{\mathbf{z}^s_i\}_{i=1,2,\cdots,n_s}$ from section s and $\{\mathbf{z}^{s+1}_i\}_{i=1,2,\cdots,n_{s+1}}$ from section $s+1$ occurs. Encoder network $f_Z(\cdot)$ is trained to compete against discriminator $D^s(\cdot)$, aiming to mix $\{\mathbf{z}^s_i\}_{i=1,2,\cdots,n_s}$ from section s and $\{\mathbf{z}^{s+1}_i\}_{i=1,2,\cdots,n_{s+1}}$ from section $s+1$. Through this competition, discriminator $D^s(\cdot)$ provides feedback to improve encoder $f_Z(\cdot)$ until representations $\{\mathbf{z}^s_i\}_{i=1,2,\cdots,n_s}$ and $\{\mathbf{z}^{s+1}_i\}_{i=1,2,\cdots,n_{s+1}}$ from sections s and $s+1$ are well-integrated. INSPIRE introduces $S-1$ discriminators $\{D^s(\cdot)\}_{i=1,2,\cdots,S-1}$ for aligning all the S tissue sections. Guided by the $S-1$ discriminators, encoder $f_Z(\cdot)$ learns to generate integrated representations of cells or spatial spots across all S tissue sections. In all experiments, we used $S-1$ discriminators, when analyzing S sections. We additionally evaluated INSPIRE's performance using C^2_S discriminators, where a separate discriminator was introduced for each pair of analyzed sections. Further details are provided in Supplementary Note 11 and Supplementary Figs. 93–96.

Based on the shared latent space, INSPIRE then achieves a harmonized NMF of gene expression across all input sections, without confounding by complex unwanted variations. The hidden spatial factors identified by this integrated NMF across sections provide a unified characterization of fine-grained tissue structures across all sections. Meanwhile, the gene-loading matrix describes gene modules associated with each spatial factor, interpreting the biological meanings of the detailed spatial organization patterns discovered by the spatial factors. We assume there are K hidden spatial factors in input sections. Each spatial factor characterizes a fine-grained spatial structure in the tissue. Let $\boldsymbol{\beta}^s_i = (\beta^s_{i,1}, \beta^s_{i,2}, \cdots, \beta^s_{i,K})$ denote the set of non-negative weights among the K hidden spatial factors for cell or spot i in section s, with $\beta^s_{i,k} \geq 0$ and $\sum_{k=1}^{K} \beta^s_{i,k} = 1$. INSPIRE generates $\boldsymbol{\beta}^s_i$ from integrated latent representation $\mathbf{z}^s_i$ of cells or spots across sections:

$$\boldsymbol{\beta}^s_i = f_\beta\left(\mathbf{z}^s_i\right), \tag{3}$$

where network $f_\beta(\cdot)$ contains a simple linear layer, followed by a softmax function. Notably, in the shared latent space, representations $\{Z^s\}_{s=1,2,\cdots,S}$ are spatially aware and free of unwanted variations. Hence, generated from Z^s, the obtained $\beta^s = (\beta^s_{i,k}) \in \mathbb{R}^{n_s \times K}$ for different sections are also spatially informed and well-integrated across sections. The integrated $\{\beta^s\}_{s=1,2,\cdots,S}$ together with shared gene loading matrix $\mu = (\mu_{k,g}) \in \mathbb{R}^{K \times G}$ across sections form a harmonized NMF model for all input sections. In the model, $\{\beta^s \mu\}_{s=1,2,\cdots,S}$ focus on capturing shared biological signal in all sections and further decomposing it into a set of K interpretable spatial factors. To account for confounding factors, including batch effects and technical effects across sections, INSPIRE also introduces section-specific and gene-specific effects $\gamma^s_g \in \mathbb{R}$ to the integrated NMF model. The detailed illustration of γ's contribution to the integration is provided in Supplementary Notes 12 and 13 and Supplementary Figs. 97–102. Combining non-negative weights for spatial factors in cells or spots $\{\beta^s\}_{s=1,2,\cdots,S}$, non-negative gene loadings μ shared across sections, and section-specific and gene-specific effects $\{\boldsymbol{\gamma}^s\}_{s=1,2,\cdots,S}$, $\boldsymbol{\gamma}^s = \left(\gamma^s_g, \gamma^s_2, \cdots, \gamma^s_G\right) \in \mathbb{R}^G$ for modeling unwanted variations, INSPIRE reconstructs the observed gene expression counts using the following integrated NMF-based model across all tissue sections:

$$y^s_{i,g} \sim \text{Poisson}\left(l^s_i u^s_{i,g}\right), \tag{4}$$

$$u^s_{i,g} = \exp\left(\log\left(\sum_{k=1}^{K} \beta^s_{i,k} \mu_{k,g}\right) + \gamma^s_g\right), \tag{5}$$

where l^s_i is the observed total transcript count in cell or spot i from section s, gene loadings satisfy $\mu_{k,g} \geq 0$ and $\sum_{g=1}^{G} \mu_{k,g} = 1$. For gene loadings μ and section-specific and gene-specific effects $\{\boldsymbol{\gamma}^s\}_{s=1,2,\cdots,S}$, INSPIRE models them as learnable parameters. After training, INSPIRE's learned $\boldsymbol{\mu}_k = (\mu_{k,1}, \mu_{k,2}, \cdots, \mu_{k,G}) \in \mathbb{R}^G$ reveals the gene signature corresponding to hidden spatial factor k, where a high value of $\mu_{k,g}$ indicates a greater impact of gene g on spatial factor k. Through learning and analyzing gene loadings μ, INSPIRE is able to find gene programs that are associated with different hidden spatial factors. Meanwhile, in the integrated NMF across sections, non-negative weights $\left\{\beta^s_{\cdot,k}\right\}_{s=1,2,\cdots,S}$, $\beta^s_{\cdot,k} = \left\{\beta^s_{i,k}\right\}_{i=1,2,\cdots,n_s}$ with spatial coordinates of cells or spots describe a spatial enrichment pattern of spatial factor k across all the S sections. Using $\{\beta^s\}_{s=1,2,\cdots,S}$, INSPIRE is capable of depicting fine-grained spatial organization structures across all sections, without being confounded by unwanted variations.

INSPIRE is a unified method that incorporates the adversarial learning mechanism for data integration with the NMF model for jointly depicting interpretable spatial structures in multiple tissue sections. We propose to train INSPIRE under the following optimization framework:

$$\min_{\{f_Z, f_\beta, \mu, \gamma\}} \max_{\{D^1, D^2, \cdots, D^{S-1}\}} \sum_{s=1}^{S-1} \mathcal{L}^s_{\text{Integration}} + \mathcal{L}_{\text{NMF}} + \lambda_{\text{AE}} \mathcal{R}_{\text{AE}} + \lambda_{\text{Geometry}} \mathcal{R}_{\text{Geometry}}, \tag{6}$$

$$\mathcal{L}^{s}_{\text{Integration}} = \mathcal{L}^{s}_{\text{Integration}}\left(f_Z, D^s\right),$$
$$\mathcal{L}_{\text{NMF}} = \mathcal{L}_{\text{NMF}}\left(f_Z, f_\beta, \mu, \gamma\right),$$
$$\mathcal{R}_{\text{AE}} = \mathcal{R}_{\text{AE}}\left(f_Z\right),$$
$$\mathcal{R}_{\text{Geometry}} = \mathcal{R}_{\text{Geometry}}\left(f_Z\right),$$

where $\mathcal{L}^{s}_{\text{Integration}}$ is the objective function of adversarial learning for integrating data from section s and section $s+1$; $\mathcal{L}_{\text{NMF}}$ is the objective function of the joint NMF for reconstructing gene expression counts across all tissue sections; $\mathcal{R}_{\text{AE}}$ and $\mathcal{R}_{\text{Geometry}}$ are regularizers to encourage the preservation of biological signals across sections in the shared latent space; λ_{AE} and $\lambda_{\text{Geometry}}$ are coefficients for the two regularizers respectively, and are set to $\lambda_{\text{AE}} = 1.0$ and $\lambda_{\text{Geometry}} = 0.02$. We explain each component in the optimization problem in equation (6) in detail in the next sections. The parameters in INSPIRE include—parameters in network $f_Z(\cdot)$ that encodes latent representations $\{Z^s\}_{s=1,2,\cdots,S}$; parameters in network $f_\beta(\cdot)$ that generates spatial factors for cells or spots among sections $\{\beta^s\}_{s=1,2,\cdots,S}$; gene-loading matrix μ shared across sections; parameters in $S-1$ discriminators $\{D^s(\cdot)\}_{s=1,2,\cdots,S-1}$ that assist data integration across sections, as well as section-specific and gene-specific effects $\gamma = \{\boldsymbol{\gamma}^s\}_{s=1,2,\cdots,S}$ that account for unwanted variations. After training, INSPIRE simultaneously outputs latent representations $\{Z^s\}_{s=1,2,\cdots,S}$, spatial factors $\{\beta^s\}_{s=1,2,\cdots,S}$ and gene loadings μ. The latent representations of cells or spatial spots are used for identifying major spatial domains in tissues and detecting spatial trajectories. Detailed spatial factors $\{\beta^s\}_{s=1,2,\cdots,S}$ are used for the discovery of fine-grained tissue subregions and spatial distributions of cell types, providing a characterization of spatial patterns in tissues at a higher resolution. Gene-loading matrix μ characterizes gene signatures associated with the detailed spatial structures discovered by spatial factors. It deciphers the biological meaning of spatial factors through factor-specific gene program detection and pathway enrichment analysis.

In INSPIRE, latent representation $\mathbf{z}_i^s$ is assumed to be deterministically generated from the gene expression, with factor $\boldsymbol{\beta}_i^s$ also deterministically derived from $\mathbf{z}_i^s$. An alternative approach would involve constructing a generative model for the gene expression levels, where the latent representation follows a Gaussian prior, and the model is solved using variational inference. However, we found this approach to be less effective than the original version of INSPIRE, both for representation learning and for spatial factor and gene program inference. Additional details are provided in Supplementary Note 14 and Supplementary Figs. 103–107.

In addition, we noted that the INSPIRE framework does not impose a fixed upper limit on the number of sections. To show INSPIRE's effectiveness in analyzing a large number of tissue sections, we applied INSPIRE to the integrative analysis of 35 coronal mouse brain sections. More details are provided in Supplementary Note 15 and Supplementary Figs. 108–110.

Next, we introduce different components of the INSPIRE framework in the following subsections. We also provide a discussion illustrating model designs and potential variants of INSPIRE in Supplementary Note 16 and Supplementary Figs. 111–129.

Adversarial learning mechanism for data integration across sections. The adversarial training between the discriminators and the encoder is formulated as a min–max optimization problem, $\min_{\{f_Z\}}\max_{\{D^1,D^2,\cdots,D^{S-1}\}}\sum_{s=1}^{S-1}\mathcal{L}^{s}_{\text{Integration}}\left(f_Z, D^s\right)$, contained in equation (6), where

$$\mathcal{L}^{s}_{\text{Integration}} = \frac{1}{n_s}\sum_{i=1}^{n_s}\log D^s\left(\mathbf{z}_i^s\right) + \frac{1}{n_{s+1}}\sum_{i=1}^{n_{s+1}}\log\left(1 - D^s\left(\mathbf{z}_i^{s+1}\right)\right).$$

The latent representations are obtained using equations (1) and (2). Given latent codes $\{\mathbf{z}_i^s\}_{i=1,2,\cdots,n_s}$ and $\{\mathbf{z}_i^{s+1}\}_{i=1,2,\cdots,n_{s+1}}$ generated by $f_Z(\cdot)$, discriminator $D^s(\cdot): \mathcal{Z} \to (0,1)$ is trained to distinguish between $\{\mathbf{z}_i^s\}_{i=1,2,\cdots,n_s}$ from section s and $\{\mathbf{z}_i^{s+1}\}_{i=1,2,\cdots,n_{s+1}}$ from section $s+1$. Here $D^s(\cdot)$ is trained to output a high score (close to one) for representations in section s, while it learns to assign a low score (close to zero) for representations in section $s+1$. This is achieved by maximizing $\mathcal{L}^{s}_{\text{Integration}}$ with respect to $D^s(\cdot)$. Given discriminators $\{D^s(\cdot)\}_{s=1,2,\cdots,S-1}$, encoder $f_Z(\cdot)$ is trained to mix latent representations across all sections, such that any discriminator cannot distinguish latent codes across sections. This is achieved by minimizing $\sum_{s=1}^{S-1}\mathcal{L}^{s}_{\text{Integration}}$ with respect to $f_Z(\cdot)$. Through the competition between encoder $f_Z(\cdot)$ and discriminators $\{D^s(\cdot)\}_{s=1,2,\cdots,S-1}$, the discriminators will guide the improvement of the encoder until the encoder generates integrated latent representations for cells or spatial spots across all the S sections.

With the above design, discriminator $D^s(\cdot)$ will guide $f_Z(\cdot)$ to mix $\{\mathbf{z}_i^s\}_{i=1,2,\cdots,n_s}$ from section s with $\{\mathbf{z}_i^{s+1}\}_{i=1,2,\cdots,n_{s+1}}$ from section $s+1$. However, a section-specific cell or spot population should not be mixed with cells or spots from another section. To preserve section-specific cell populations from being incorrectly mixed with other cells, we follow our previous work[36] to adopt a threshold for discriminator scores. Consider a cell population that is unique in section s. Discriminator $D^s(\cdot)$ can easily recognize cells in this population as cells from section s, and assign extremely high scores to them. By similar reasoning, $D^s(\cdot)$ will assign extremely low scores to section $s+1$-unique cell populations. Therefore, as section-unique cell populations are prone to be assigned with extreme discriminator scores, we set boundaries for discriminator scores to make discriminators inactive on them. Specifically, the outputs of standard discriminators are transformed into $(0, 1)$ through the sigmoid function. For any applicable latent code $\mathbf{z}$, $D^s(\mathbf{z}) = \text{sigmoid}\left(d^s(\mathbf{z})\right) = 1/\left(1+\exp\left(-d^s(\mathbf{z})\right)\right)$, where $d^s(\mathbf{z}) \in \mathbb{R}$ is the logit value of discriminator score $D^s(z)$. We bound discriminator score $D^s(\mathbf{z}))$ by thresholding its logit $d^s(\mathbf{z})$ to a reasonable range $(-m, m)$, where m is set to 5.0:

$$\widetilde{D}^s(\mathbf{z}) = \text{sigmoid}\left(\text{clamp}\left(d^s(\mathbf{z})\right)\right),$$

where $\text{clamp}(\cdot) = \max\left(\min(\cdot, m), -m\right)$, $m > 0$. By clamping $d^s(\mathbf{z})$, $\widetilde{D}^s(\mathbf{z})$ becomes a constant when $d^s(\mathbf{z}) < -m$ or $d^s(\mathbf{z}) > m$, providing zero gradients for updating parameters in encoder network $f_Z(\cdot)$. With this design, the section-unique cell populations with extreme $d^s(\mathbf{z})$ scores will be left in the inactive region of discriminators. Consequently, discriminators will not force encoder $f_Z(\cdot)$ to mix section-unique cell populations with other cells, avoiding incorrect integration. Meanwhile, $\widetilde{D}^s(\mathbf{z})$ remains the same as $D^s(\mathbf{z})$ when $\widetilde{D}^s(\mathbf{z}) \in (-m, m)$, effectively guiding encoder $f_Z(\cdot)$ to align cells that are likely to belong to the shared cell populations among sections for data integration. For clarity, we still use the notation $D^s(\cdot)$ to denote discriminator $\widetilde{D}^s(\cdot)$ with the score thresholding design hereinafter. In the sensitivity analysis, we showed that INSPIRE's results were relatively robust to variations in the choice of m. Details are provided in Supplementary Note 2 and Supplementary Figs. 28–30.

Joint NMF for reconstructing gene expressions in multiple ST sections. The major objective function in optimization problem in equation (6) of INSPIRE, $\mathcal{L}_{\text{NMF}}$, corresponds to the reconstruction of gene expression counts in all S input sections through a joint NMF model. INSPIRE uses encoder $f_Z(\cdot)$ to generate integrated data $\{Z^s\}_{s=1,2,\cdots,S}$ across sections, guided by discriminators $\{D^s(\cdot)\}_{s=1,2,\cdots,S-1}$. By equation (3), it then leverages network $f_\beta(\cdot)$ to decompose the signal captured in $\{Z^s\}_{s=1,2,\cdots,S}$ into a set of K interpretable spatial factors $\{\beta\}^s_{s=1,2,\cdots,S}$ that characterize spatial structures of tissues at a fine-grained level. Combining the obtained spatial factors with additional parameters, including gene loadings μ as well as section-specific and gene-specific effects γ, INSPIRE reconstructs the gene expression counts from all input sections through an integrated NMF-based model described by equations (4) and (5). Based on this model, the corresponding objective function is given by

$$\mathcal{L}_{\mathrm{NMF}} = -\sum_{s=1}^{S}\frac{1}{n_s}\sum_{i=1}^{n_s}\sum_{g=1}^{G}\left(y_{i,g}^s \log\left(l_i^s u_{i,g}^s\right) - l_i^s u_{i,g}^s\right).$$

By minimizing $\mathcal{L}_{\mathrm{NMF}}$ with respect to $f_Z(\cdot)$, $f_\beta(\cdot)$, μ and γ, INSPIRE deciphers interpretable hidden spatial factors that are unified across sections with $\{\beta^s\mu\}$. Here $\{\beta^s\}_{s=1,2,\cdots,S}$ characterizes detailed spatial organizations in tissues, while μ describes gene signatures associated with the tissue organization patterns identified by spatial factors for interpretability.

Regularization for encouraging the preservation of biological variations across sections. INSPIRE uses two regularizers, $\mathcal{R}_{\mathrm{AE}}$ and $\mathcal{R}_{\mathrm{Geometry}}$, to help preserve biological signals across sections in the shared latent space. We design regularizer $\mathcal{R}_{\mathrm{AE}}$ as

$$\mathcal{R}_{\mathrm{AE}} = \sum_{s=1}^{S}\frac{1}{n_s}\sum_{i=1}^{n_s}\| \mathbf{x}_i^s - f_X^s(\mathbf{z}_i^s, s)\|_2,$$

where section-specific neural network $f_X^s(\cdot,s)$ is introduced to reconstruct log-normalized gene expressions in cells or spots $\mathbf{x}_i^s$ based on latent codes $\mathbf{z}_i^s$ and section label s. Encoder $f_Z(\cdot)$ and network $f_X^s(\cdot,s)$ together form an auto-encoder structure between log-normalized data $\{X^s\}_{s=1,2,\cdots,S}$ and latent codes $\{Z^s\}_{s=1,2,\cdots,S}$. In regularizer $\mathcal{R}_{\mathrm{AE}}$, section-specific network $f_X^s(\cdot,s)$ is designed to recover gene expressions from the latent space while accounting for section-specific effects using section labels. Therefore, encoder $f_Z(\cdot)$ is encouraged to distill all biological information into the latent space without section-specific effects. To preserve a good geometric structure in the latent space for revealing biological signals, for example, continued developmental trajectories of cells, we propose regularizer $\mathcal{R}_{\mathrm{Geometry}}$:

$$\begin{aligned} R_{\mathrm{Geometry}} &= \sum_{s=1}^{S}\frac{1}{n_s}\sum_{i=1}^{n_s}\| c_{x,i}^s - c_{z,i}^s\|_2^2,\\ c_{x,i,j}^s &= 1 - \frac{<\mathbf{x}_i^s, \mathbf{x}_j^s>}{\|\mathbf{x}_i^s\|_2 \, \|\mathbf{x}_j^s\|_2},\\ c_{z,i,j}^s &= 1 - \frac{<\mathbf{z}_i^s, \mathbf{z}_j^s>}{\|\mathbf{z}_i^s\|_2 \, \|\mathbf{z}_j^s\|_2}, \end{aligned}$$

where for cells or spots i and j from section s, $c_{x,i,j}^s$ is the cosine similarity between their log-normalized gene expressions $\mathbf{x}_i^s$ and $\mathbf{x}_j^s$, $c_{z,i,j}^s$ is the cosine similarity between their latent representations $\mathbf{z}_i^s$ and $\mathbf{z}_j^s$; $c_x^s = (c_{x,i,j}^s) \in \mathbb{R}^{n_s\times n_s}$ and $c_z^s = (c_{z,i,j}^s) \in \mathbb{R}^{n_s\times n_s}$ are corresponding cosine similarity matrices; $c_{x,i}^s$ and $c_{x,i}^s$ are the i-th rows of c_x^s and c_z^s, respectively. In regularizer $\mathcal{R}_{\mathrm{Geometry}}$, cells or spots with high similarities in gene expressions are encouraged to be close in the latent space, while cells or spots with dissimilar gene expressions are encouraged to remain distinct in the latent space. Hence, by using $\mathcal{R}_{\mathrm{Geometry}}$, INSPIRE encourages $f_Z(\cdot)$ to preserve biological meaningful structures in the shared latent space.

Selection of informative genes

When all input sections provide the whole transcriptome profiling, INSPIRE selects the informative genes and uses them as features. Following the Scanpy pipeline[76], INSPIRE selects the top M highly variable genes for each section. It then takes the intersection of these highly variable genes across all sections to ensure that the features of cells or spatial spots are shared across all sections. By default, we set $M = 6{,}000$. If the number of selected features is less than 2,000 with $M = 6{,}000$, a larger value of M can be adopted. When some sections to be analyzed are based on ST technologies that measure the expressions of a limited number of genes, such as MERFISH, INSPIRE uses all the shared genes across the input sections for the integrative analysis.

Network structures

Encoder $f_Z(\cdot)$ contains a GNN layer and a dense layer. The GNN layer takes log-normalized gene expressions $\mathbf{x}_i^s$ and spatial graph A^s as input. It outputs 512-dimensional hidden vectors. Then the dense layer in $f_Z(\cdot)$ maps the 512-dimensional hidden vectors to the P-dimensional latent representations $\mathbf{z}_i^s$ of cells or spatial spots. We set $P = 32$ throughout all analyses. Inspired by previous works[17,40,46,77,78], INSPIRE provides two options for the GNN layer—the graph attention layer and the lightweight graph-convolutional layer. The graph attention layer is formulated as:

$$\begin{aligned} \mathbf{h}_i^s &= \sigma\left(\sum_{j\in\varepsilon_i}\alpha_{i,j}W\mathbf{x}_i^s\right),\\ \alpha_{i,j}^s &= \frac{\exp(e_{i,j}^s)}{\sum_{j'\in\varepsilon_i}\exp(e_{i,j'}^s)},\\ e_{i,j}^s &= \mathrm{sigmoid}\left(\mathbf{v}_1^{\mathrm{T}}W\mathbf{x}_i^s + \mathbf{v}_2^{\mathrm{T}}W\mathbf{x}_j^s\right), \end{aligned}$$

where $\mathbf{h}_i^s$ represents the output of the graph attention layer; $\varepsilon_i = \{j|a_{i,j}^s = 1\}$ represents the neighbor set of cell or spot i encoded in spatial graph A^s; W, $\mathbf{v}_1$ and $\mathbf{v}_2$ are parameters in the graph attention layer; $\sigma(\cdot)$ is the activation function. Based on the graph attention mechanism, parameters $\mathbf{v}_1$ and $\mathbf{v}_2$ are used to learn edge weights $\alpha_{i,j}^s$ between neighboring cells and spots, helping to adaptively borrow information from neighboring cells or spots. INSPIRE adopts the graph attention layer in $f_Z(\cdot)$ when handling the integration task for ST datasets with moderate numbers of cells or spatial spots. To integrate large atlas-scale ST datasets that contain hundreds of thousands or even millions of cells or spots, INSPIRE uses the lightweight graph-convolutional layer in $f_Z(\cdot)$, which is formulated as:

$$\begin{aligned} H^s &= \sigma\left(W\widetilde{X}^s\right),\\ \widetilde{X}^s &= \mathrm{concat}\left(X^s, \widetilde{A}^s X^s, (\widetilde{A}^s)^2 X^s, \cdots, (\widetilde{A}^s)^L X^s\right), \end{aligned}$$

where H^s represents the output of the lightweight graph-convolutional layer; W denotes the parameters; $\widetilde{A}^s = (\widetilde{D}^s)^{-1/2}A^s(\widetilde{D}^s)^{-1/2}$, $\widetilde{D}^s$ is the diagonal degree matrix of A^s and L is the number of steps in the concatenation. We set $L = 1$ by default. The graph attention layer has the advantage of providing an inference of the edge importance between neighborhood cells and spots for adaptively aggregating information in microenvironments of cells or spots. By preparing $\widetilde{X}^s$ as a preprocessing step, the lightweight graph-convolutional layer enables efficient training with mini-batch samples from $\widetilde{X}^s$, serving as a scalable approach to account for spatial dependencies in datasets with large numbers of cells or spots. A comparison illustrating the trade-off between the usage of graph attention networks and lightweight graph-convolutional networks within the INSPIRE framework can be found in Supplementary Note 17 and Supplementary Fig. 130.

For network $f_\beta(\cdot)$ that produces K-dimensional spatial factors $\boldsymbol{\beta}_i^s$ from latent codes $\mathbf{z}_i^s$ of cells or spatial spots, INSPIRE adopts a one-layer dense network with the softmax activation. The choice of K depends on the scale of the tissues to be analyzed. For example, INSPIRE set $K = 20$ for analyzing the cortex region of the brain, while it adopts $K = 40$ for analyzing the whole-brain sections.

For discriminators $\{D^s(\cdot)\}_{s=1,2,\cdots,S-1}$ that guide encoder $f_Z(\cdot)$ to achieve the data integration, INSPIRE uses two-layer dense networks. $D^s(\cdot)$ works as a binary classifier to distinguish between $\{\mathbf{z}_i^s\}_{i=1,2,\cdots,n_s}$ from section s and $\{\mathbf{z}_i^{s+1}\}_{i=1,2,\cdots,n_{s+1}}$ from section s. Then, encoder $f_Z(\cdot)$

https://doi.org/10.1038/s41588-026-02579-x

competes against $D^s(\cdot)$ to integrate $\{\mathbf{z}_i^s\}_{i=1,2,\cdots,n_s}$ with $\{\mathbf{z}_i^{s+1}\}_{i=1,2,\cdots,n_{s+1}}$. Here $D^s(\cdot)$ takes latent codes z of cells or spots from sections s or $s+1$ as input. It first uses a dense layer with an activation function to map z to a 512-dimensional hidden state. It then uses another dense layer to produce a score belonging to (0, 1) from the 512-dimensional hidden state.

INSPIRE introduces section-specific network $f_X^s(\cdot,s)$ in its regularizer $\mathcal{R}_{AE}$ to help preserve biological variations across sections in the shared latent space. $f_X^s(\cdot,s)$ takes latent representations $\mathbf{z}_i^s$ and section label s as inputs. It adopts a graph attention layer or a lightweight graph-convolutional layer, followed by a dense layer, to recover log-normalized gene expressions $\mathbf{x}_i^s$ while accounting for section-specific effects and spatial dependencies among cells or spatial spots. The dimensionality of the hidden state in $f_X^s(\cdot,s)$ is set to be 512.

Model training details

INSPIRE uses Adamax, which is a variant of the Adam algorithm[79], for stochastic optimization during model training. By default, the number of optimization steps in INSPIRE is set to 10,000 with learning rate lr = 0.0005, coefficients for computing running averages $\beta_1 = 0.9$, $\beta_2 = 0.999$ and weight decay parameter $\lambda = 0.0001$. We conducted all experiments on a single graphics processing unit. The computation times for all experiments are detailed in Supplementary Table 1.

Evaluation metrics

We evaluated spot or cell representations using ASW and assessed spatial domain identification results, inferred from these representations, with ARI and NMI metrics. The quality of spatial factors was measured by factor diversity and factor coherence. The integration performance was evaluated by iLISI and cLISI metrics. Spatial domain continuity was assessed using the spatial discontinuity metric.

ASW. ASW calculates the silhouette width of cells or spatial spots with respect to spatial region annotation labels. The silhouette width for cell or spot i, assigned to annotated cluster C, is defined as $s(i) = \frac{b(i)-a(i)}{\max\{a(i),b(i)\}}$, where $a(i)$ denotes the average distance between cell i and all other cells within the same cluster C, and $b(i)$ denotes the minimum value of average distance from cell i to all cells in each cluster other than cluster C. The ASW is then computed as the mean of $s(i)$ across all cells or spots. Higher ASW scores indicate better separation between clusters and greater cohesion within clusters, reflecting improved preservation of clustering structure in the integrated representations.

ARI. ARI measures the alignment between the spatial domain identification result and the expert manual annotation. A lower score suggests that the two sets of labels for cells or spots are independent, while a higher score indicates that the labels are identical, except for a possible permutation.

NMI. NMI is calculated between the spatial domain identification result and the expert manual annotation. A low NMI value indicates minimal shared information between the label sets, while a high value suggests a strong correlation between them.

Factor diversity. Factor diversity is defined as the percentage of unique genes among the top ten genes across all factors. A higher score reflects greater diversity among factors, while a lower score indicates more redundancy.

Factor coherence. Factor coherence measures the interpretability of factors by calculating the average pointwise mutual information of top genes associated with each factor, then averaging these values across all factors. Specifically,

$$\text{Factorcoherence} = \frac{1}{K}\sum_{k=1}^{K}\frac{1}{45}\sum_{i=1}^{10}\sum_{j=i+1}^{10} m(g_i^{(k)}, g_j^{(k)}),$$

$$m(g_i, g_j) = -\frac{\log\frac{P(g_i, g_j)}{P(g_i)P(g_j)}}{\log P(g_i, g_j)},$$

where $\{g_1^{(k)}, g_2^{(k)}, \cdots, g_{10}^{(k)}\}$ denotes the top 10 genes associated with factor k, $m(\cdot,\cdot)$ is the normalized pointwise mutual information, $P(g_i, g_j)$ is the probability of genes g_i and g_j coexpressing in a spot or cell and $P(g_i)$ is the marginal probability of gene g_i.

iLISI. The iLISI score evaluates the effectiveness of data alignment across datasets. It is defined as the effective number of datasets in a neighborhood, where 1 indicates poor mixing and 2 indicates good mixing between two datasets. Higher values indicate good data alignment performance.

cLISI. The cLISI score evaluates biological variation preservation performance. It measures the effective number of cell types in a neighborhood, where 1 indicates that the cell population is well preserved, and larger values indicate greater mixing of different cell populations. Lower values indicate good performance in preserving biological variation.

Spatial discontinuity. The spatial discontinuity score quantifies the proportion of spatial spots that lack any neighboring spot assigned to the same spatial region.

Rankings of genes within a spatial factor

The gene loading associated with spatial factor k is represented as $\boldsymbol{\mu}_{k,\cdot} = (\mu_{k,1}, \mu_{k,2}, \cdots, \mu_{k,G})$, where μ is the gene-loading matrix derived from integrated NMF across all input sections, and G is the total number of genes analyzed. The non-negative values in $\boldsymbol{\mu}_{k,\cdot}$ indicate the relative expression levels among genes within a factor, with a higher $\mu_{k,g}$ corresponding to greater enrichment of gene g expression in factor k. Therefore, for each spatial factor k, we can rank genes according to the non-negative values in $\boldsymbol{\mu}_{k,\cdot}$. The gene with the highest value is ranked first, and the gene with the lowest value is ranked last.

Identification of genes specific to a spatial factor

To identify highly expressed genes specific to spatial factor k, we first select the top G_0-ranked genes based on the gene loading associated with spatial factor k, with G_0 set to 50 by default. For each selected gene g, we then calculate the fold change between its estimated weight on spatial factor k and all other spatial factors. A gene g is considered highly expressed and specific to spatial factor k, if $\mu_{k,g}/\mu_{k',g} > 1$ for any $k' \neq k$.

Number of spatial factors

In the human DLPFC example, we demonstrated that the NMF component in INSPIRE enhances the accuracy of spot representations in the latent space. In addition, we investigated how varying the number of spatial factors in INSPIRE affects the quality of these spot representations. The results indicate that the quality remained stable across different numbers of spatial factors, with a slight improvement in accuracy as the number of spatial factors increased (Supplementary Note 18 and Supplementary Fig. 131).

Next, we examined the relationship between the number of spatial factors and both the quality of the spatial factors and the model fitting accuracy using the human DLPFC and mouse brain data. The results suggest that increasing the number of spatial factors not only improves model fit to the ST datasets but also reduces the diversity of the spatial factors. These examples illustrate that the optimal number of spatial factors depends on the scale of the ST sections. Specifically,

 https://doi.org/10.1038/s41588-026-02579-x

for the human DLPFC sections, which represent only a subregion of the brain, the factor diversity score dropped below 50% when the number of spatial factors exceeded 20. In contrast, for the mouse brain data, which includes multiple complementary views of the brain, the factor diversity score remained above 50% even with 40 spatial factors (Supplementary Note 19 and Supplementary Figs. 132 and 133). Based on empirical observations, we recommend using number of spatial factors $K = 20$ for analyzing a subregion of an organ, $K = 40$ for a complex organ and $K = 60$ for a whole organism. Alternatively, the INSPIRE model can be run with different values of spatial factor number K, and we recommend selecting the largest K such that factor diversity exceeds a specified threshold. By default, we suggest a threshold of 50%.

For other spatially aware NMF-based methods, we applied the number of spatial factors consistent with that used in INSPIRE's analyses to ensure that each method captured the data structure within subspaces of equal dimensionality, enabling a fair comparison. We further examined the robustness of INSPIRE to the choice of K and compared its performance with other spatially aware NMF-based methods, SpiceMix and NSFH, across a range of K values. We applied INSPIRE, SpiceMix and NSFH to DLPFC sections 151673–151676. INSPIRE demonstrated relatively stable performance in spatial domain identification and preservation of layer structures, as quantified by ARI, NMI and ASW scores. Across all tested values of K, INSPIRE outperformed both SpiceMix and NSFH in these metrics (Supplementary Fig. 134). In addition, INSPIRE consistently achieved higher scores on both factor diversity and factor coherence metrics than the competing methods, further demonstrating its superior quality of spatial factor inference across different values of K (Supplementary Fig. 135). We then examined the impact of K on integration performance using three Visium mouse brain sections with partially overlapping spatial structures. SpiceMix and NSFH, which do not incorporate mechanisms to correct for batch effects, exhibited limited alignment of shared structures such as the isocortex and hippocampus across sections, consistently across different choices of K (Supplementary Figs. 136–138). While INSPIRE achieved good performance robust to the choices of K, as indicated by its consistently high iLISI scores compared to other methods (Supplementary Fig. 139). More details are provided in Supplementary Note 20.

Additional benchmarking results

Using DLPFC sections with labels 151673–151676, we additionally performed joint analyses on two sections at a time, including an integrative analysis of sections 151673 and 151674, as well as an integrative analysis of sections 151675 and 151676. INSPIRE's four-section analysis consistently outperformed its two-section analysis, demonstrating its ability to effectively leverage information across multiple sections. Moreover, INSPIRE's two-section analysis achieved superior spatial domain identification compared to all other methods. More details are provided in Supplementary Note 21 and Supplementary Figs. 140–143.

Running time and peak memory usage

We conducted a comparative analysis of runtime and memory usage for INSPIRE and two other spatially aware NMF-based methods, SpiceMix and NSFH, all of which are designed to infer gene programs from ST data. For this evaluation, we used two high-resolution human breast cancer tissue sections profiled using the Xenium platform, comprising 283,063 segmented cells in total, with expression data for 313 genes. To assess scalability, we generated datasets of varying sizes by subsampling cells from each section, with sample sizes ranging from 10,000 to the full 283,063 cells. All methods were run on the same hardware, a GPU node equipped with an NVIDIA RTX 5000. SpiceMix and NSFH were executed until their GPU memory usage exceeded the 16 GB limit of the RTX 5000, while INSPIRE was successfully applied to all datasets. In this comparison, we found that, among spatially aware NMF-based methods, SpiceMix required substantially longer runtime compared to NSFH and INSPIRE, and both SpiceMix and NSFH exhibited high GPU memory usage. In contrast, INSPIRE demonstrated strong scalability and relative efficiency in terms of runtime and GPU memory consumption. These results highlight INSPIRE's computational efficiency and memory scalability for large-scale ST datasets. Details are provided in Supplementary Note 7 and Supplementary Fig. 85.

To better evaluate INSPIRE's scalability, we additionally conducted experiments to record INSPIRE's running time and peak memory usage with large-scale synthetic ST datasets. Specifically, we created two baseline sections, each containing 500,000 cells and 480 genes. We varied the number of cells per section from 500,000 to 2 million, resulting in a total cell count ranging from 1 million to 4 million, to evaluate the computational consumption with respect to the total number of cells. Then, we fixed the number of cells per section at 500,000 and increased the number of sections from two (1 million cells in total) to ten (5 million cells in total) to evaluate the scalability of INSPIRE with respect to both the number of analyzed sections and the number of cells. Next, we evaluated the scalability of INSPIRE with respect to the number of genes (varied from 480 to 3,000), while fixing the number of cells per section at 500,000 and the number of sections at two. Finally, we used two synthetic sections, each containing 500,000 cells and 480 genes, and varied the number of spatial factors from 10 to 60 to assess its impact on computational performance. INSPIRE was successfully applied to all of the created large-scale data analysis scenarios using a GPU node with GTX 5000, showing its scalability. Its running time, CPU and GPU memory, either remained stable or showed a linear increase with respect to the variations. Details are provided in Supplementary Note 8 and Supplementary Figs. 86–89.

Non-NMF version of the INSPIRE framework

In the human DLPFC section analysis, we manually removed the NMF component from INSPIRE. Comparison of results from INSPIRE with and without NMF shows that the NMF component in INSPIRE serves as an additional form of regularization, which guides the latent representations in INSPIRE to capture biologically meaningful signals while avoiding overfitting to noise, thereby gaining accuracy in the latent representations (Supplementary Fig. 18).

To remove the NMF component from INSPIRE, we excluded the NMF loss term $\mathcal{L}_{\mathrm{NMF}}$ from the training objective. The modified optimization problem is formulated as:

$$\min_{\{f_Z, f_X, f_\beta\}} \max_{\{D^1, D^2, \cdots, D^{S-1}\}} \sum_{s=1}^{S-1} \mathcal{L}^s_{\mathrm{Integration}} + \lambda_{\mathrm{AE}}\, \mathcal{R}_{\mathrm{AE}} + \lambda_{\mathrm{Geometry}}\, \mathcal{R}_{\mathrm{Geometry}}, .$$

In this setting, the latent representations are optimized primarily to reconstruct the gene expression profiles through the auto-encoder regularization term $\mathcal{R}_{\mathrm{AE}}$, and to preserve biologically meaningful structure through the geometric regularization term $\mathcal{R}_{\mathrm{Geometry}}$, while enabling cross-section alignment through the integration loss $\mathcal{L}^s_{\mathrm{Integration}}$.

Statistics and reproducibility

All data used in this study are publicly available from online sources, as described in the Data Availability section. No statistical method was used to predetermine sample size, and no data were excluded from the analyses unless explicitly stated. All code required to reproduce the results is publicly available on GitHub (https://github.com/jiazhao97/INSPIRE) and Zenodo (https://doi.org/10.5281/zenodo.18330972)[81].

Human DLPFC dataset profiled by Visium. This human brain dataset includes 47,329 spatial spots and 33,525 genes across 12 sections from three donors. Within each section, spatial spots with fewer than 50 total counts were excluded, leaving 47,322 spots for analysis.

Mouse brain sagittal anterior, sagittal posterior and coronal sections profiled by Visium. These three mouse brain datasets contain 2,825, 3,293 and 2,702 spatial spots, respectively, with 31,040, 31,040 and 14,801 genes. After filtering spatial spots with fewer than 50 total counts, no spots were removed.

Mouse brain sections profiled by MERFISH and Slide-seqV2. The mouse brain MERFISH dataset contains 44,959 cells and 1,122 genes, while the Slide-seqV2 section includes 53,208 spatial spots and 23,264 genes. Analyses were restricted to genes shared between the two datasets, resulting in 1,077 genes. Within each section, cells or spatial spots with fewer than six total counts were filtered, yielding 44,950 cells for the MERFISH section and 29,866 spatial spots for the Slide-seqV2 section.

Mouse whole-embryo sections profiled by seqFISH and Stereo-seq. The whole-embryo seqFISH section contains 14,185 cells and 351 genes, while the Stereo-seq section includes 5,913 spatial spots and 25,568 genes. Analyses were restricted to genes shared between the two datasets, yielding 347 genes. Within each section, cells or spatial spots with fewer than two total counts were filtered. No cells were removed from the seqFISH section, and 5,880 spatial spots were retained for analysis in the Stereo-seq section.

Mouse skin sections during wound healing profiled by Visium. The uninjured mouse skin section and the postoperative day 7 wound section contain 949 and 1,671 spatial spots, respectively, with 32,272 genes in each section. Within each section, spatial spots with fewer than 20 total counts were filtered. No spots were removed from the postoperative day 7 wound section, and 948 spatial spots were retained for analysis in the uninjured section.

Human breast cancer sections profiled by Xenium. The two Xenium human breast cancer sections contain 167,780 and 118,752 cells, respectively, with 313 genes in each section. Within each section, cells with fewer than six total counts were filtered, yielding 164,596 and 118,467 cells for the two sections, respectively.

Mouse whole-embryo sections across developmental time points profiled by Stereo-seq. We analyzed eight Stereo-seq mouse-embryo sections, spanning E9.5 to E16.5. These sections contain 514,804 spatial spots in total. Each section has over 220,000 genes. Within each section, spatial spots with fewer than 50 total counts were excluded, leaving 514,415 spots for analysis.

Mouse hypothalamic preoptic region section profiled by MERFISH. The two MERFISH sections contain 5,557 and 5,488 cells, respectively, with 155 genes. After filtering cells with zero detected counts, no cells were removed.

Mouse hippocampus region section profiled by STARmap PLUS. The two STARmap PLUS sections contain 8,202 and 8,186 cells, respectively, with 2,766 genes. After filtering cells with zero detected counts, no cells were removed.

Mouse whole-embryo sections profiled by Stereo-seq for 3D reconstruction. Five adjacent Stereo-seq mouse-embryo sections were analyzed for 3D reconstruction of the mouse embryo, comprising a total of 568,437 spatial spots. Each section contains over 270,000 genes. Within each section, spatial spots with fewer than 50 total counts were excluded, leaving 567,381 spatial spots for analysis.

Mouse brain sections profiled by the ST platform. We analyzed 35 mouse brain ST sections comprising a total of 17,086 spatial spots and 23,371 genes. After filtering spatial spots with zero detected counts, no spots were removed.

Reporting summary
Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability
All data used in this work are publicly available through online sources. • Human DLPFC dataset profiled by the Visium platform[50] (https://research.libd.org/spatialLIBD/). • Mouse brain sagittal anterior, sagittal posterior and coronal sections profiled by Visium[32–34] (https://www.10xgenomics.com/datasets). • Mouse brain section profiled by MERFISH[58] (https://doi.org/10.35077/act-bag). • Mouse brain section profiled by Slide-seqV2 (ref. 7; https://singlecell.broadinstitute.org/single_cell). • Mouse whole-embryo section profiled by seqFISH[35] (https://crukci.shinyapps.io/SpatialMouseAtlas/). • Mouse skin sections during wound healing profiled by the Visium platform[60] (GSE178758). • Human breast cancer sections profiled by Xenium[15]((https://www.10xgenomics.com/products/xenium-in-situ/preview-dataset-human-breast). • Mouse whole-embryo datasets across different developmental time points profiled by Stereo-seq[8] (https://db.cngb.org/stomics/mosta/). • Mouse hypothalamic preoptic region section profiled by MERFISH[72] (https://doi.org/10.5061/dryad.8t8s248). • Mouse hippocampus region section profiled by STARmap PLUS[14] (https://doi.org/10.5281/zenodo.7458952). • Mouse brain sections profiled by the ST platform[80] (GSE147747). Source data are provided with this paper.

Code availability
The INSPIRE software is available at https://github.com/jiazhao97/INSPIRE. All codes are available via the Zenodo repository at https://doi.org/10.5281/zenodo.18330972 (ref. 81). For the benchmarking study among the compared methods, the following versions were used: SpiceMix (v1.0.0), NSFH (v1.0.0), PRECAST (v1.6.4), MEFISTO (v0.7.2), GraphST (v1.0.0), Harmony (v0.1.0), Seurat (v4.3.0), LIGER (v1.0.1), BASS (v1.1.0), BayesSpace (v1.8.2), BANKSY (v1.5.3), SpaGCN (v1.2.7), SpatialGlue (v1.0.0), STAGATE (v1.0.0) and SpatialLeiden (v0.2.0).

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Acknowledgements
This work was supported in part by the National Institutes of Health (grants U01 HG013840, U24 HG012108 and P50 CA196530 to J.Z. and H.Z.), NCCIH (grant R01 AT012041), the Allen Discovery Center program, and a Paul G. Allen Frontiers Group-advised program of the Paul G. Allen Family Foundation (to R.B.C.).

Author contributions
J.Z. conceived the idea and developed the method. H.Z. supervised the project. J.Z., X.Z., G.W., Y.L., T.L., R.B.C. and H.Z. designed the

https://doi.org/10.1038/s41588-026-02579-x

experiments and performed the analyses. J.Z. and H.Z. wrote the paper.

Competing interests

All authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41588-026-02579-x.

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Peer review information *Nature Genetics* thanks Nigel Chou and the other, anonymous, reviewer(s) for their contribution to the peer review of this work. Peer reviewer reports are available.

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Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	No software was used for the data collection.
Data analysis	INSPIRE (https://github.com/jiazhao97/INSPIRE, https://inspire-tutorial.readthedocs.io/en/latest), python v3.8.15, pytorch v1.12.1, pandas v1.5.3, numpy v1.24.4 were used for designing and running the algorithm. SpiceMix v1.0.0 (https://github.com/ma-compbio/SpiceMix), NSFH v1.0.0 (https://github.com/willtownes/nsf-paper), PRECAST v1.6.4 (https://github.com/feiyoung/PRECAST), MEFISTO v0.7.2 (https://github.com/bioFAM/mofapy2), GraphST v1.0.0 (https://github.com/JinmiaoChenLab/GraphST), Harmony v0.1.0 (https://github.com/immunogenomics/harmony), Seurat v4.3.0 (https://github.com/satijalab/seurat), LIGER v1.0.1 (https://github.com/welch-lab/liger), BASS v1.1.0 (https://github.com/zhengli09/BASS), BayesSpace v1.8.2 (https://github.com/edward130603/BayesSpace), BANKSY v1.5.3 (https://github.com/prabhakarlab/Banksy), SpaGCN v1.2.7 (https://github.com/jianhuupenn/SpaGCN), SpatialGlue v1.0.0 (https://github.com/JinmiaoChenLab/SpatialGlue), STAGATE v1.0.0 (https://github.com/zhanglabtools/STAGATE), SpatialLeiden v0.2.0 (https://github.com/HiDiHlabs/SpatialLeiden) were used for the benchmarking study. scanpy v1.8.2, anndata v0.9.2, louvain v0.8.0, umap-learn v0.5.5, matplotlib v3.5.3 were used for preprocessing and post-processing of the spatial transcriptomics data.

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All data used in this work are publicly available through online sources.
Human dorsolateral prefrontal cortex dataset profiled by Visium platform (https://research.libd.org/spatialLIBD/);
Mouse brain sagittal anterior, sagittal posterior, and coronal sections profiled by Visium (https://www.10xgenomics.com/datasets);
Mouse brain section profiled by MERFISH (https://doi.brainimagelibrary.org/doi/10.35077/act-bag);
Mouse brain section profiled by Slide-seq V2 (https://singlecell.broadinstitute.org/single_cell);
Mouse whole-embryo section profiled by seqFISH (https://crukci.shinyapps.io/SpatialMouseAtlas/);
Mouse whole-embryo datasets across different developmental time points profiled by Stereo-seq (https://db.cngb.org/stomics/mosta/);
Mouse hypothalamic preoptic region sections profiled by MERFISH (https://doi.org/10.5061/dryad.8t8s248);
Mouse hippocampus region sections profiled by SRARmap PLUS (https://doi.org/10.5281/zenodo.7458952);
Mouse skin sections during wound healing profiled by Visium (GSE178758);
Human breast cancer sections profiled by Xenium (https://www.10xgenomics.com/products/xenium-in-situ/preview-dataset-human-breast);
Mouse brain sections profiled by the ST platform (GSE147747).

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Sample size	We used 11 publicly available data in the manuscript. All data used in this study are publicly available from online sources. No statistical method was used to predetermine sample size.
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